TOPIC 3 · CONSERVATION LAWS

Solved Problems: Official Collection CL1 to CL16

Worked solutions to the conservation-laws collection of the Department of Thermal and Fluid Engineering, Universidad Carlos III de Madrid (16 of the 22 problems so far; the rest of the collection is added in batches). Every problem restates the official statement, then follows four phases: (1) hypotheses, data and unknowns; (2) frames, control volume and its surfaces; (3) the balances; (4) interpretation, limits and dimensional check. The balances follow the method used in class: each one is written in full with the vanishing terms crossed out and the reason underneath, every surface integral is then evaluated surface by surface (the vertical line drawn under each integral), the wall terms are grouped (boxed) as the force on the body, and each result is boxed and numbered. Discrepancies found in the handwritten official solutions are flagged in the corresponding problem, and anything beyond the course material is marked as an extension. References to Notes.pdf are to Chapter 3 (Conservation laws).

PROBLEM CL1

Airfoil Wake: Streamline Displacement and Drag

Original statement · CL1.pdf

Control volume | Linear momentum | Wake | Drag

OFFICIAL PROBLEM STATEMENT

The experimental tests of an airfoil in a wind tunnel indicate that the velocity profile upstream remains uniform, whereas behind the airfoil there exists a wake region with significant velocity deficit whose thickness increases downstream, resulting in the flow field sketched in the figure of CL1.pdf. For the airfoil tested, the velocity measurements at a given downstream section reveal that the velocity in the wake varies linearly from the value $u = U/2$ at the centerline to reach the value $u = U$ at the edge of the wake, located at $y = y_0$. To analyze the problem, consider a control volume limited laterally by the streamlines that extend from $x \to -\infty$, $y = \pm y_{-\infty}$ ($y_{-\infty} < y_0$) to $x = x_0$, $y = \pm y_0$ and internally by the surface of the airfoil. Assuming that the air behaves as an incompressible fluid with density ρ and that viscous stresses are negligible both in the wake and along the bounding streamlines, obtain the value of $y_{-\infty}$ as a function of y_0, as well as the value of the drag on the airfoil as a function of ρ, y_0 and U. In the analysis, assume that on the outer boundary of the control volume the pressure differences from the ambient value p_∞ are negligible, and so is the effect of gravity on the air flow.

Source: CL1.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eqs. 3.10-3.11, integral momentum balance Eqs. 3.35-3.36, force of the fluid on a body Eq. 3.32.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, planar (everything per unit span), incompressible (ρ constant); neglect $\bar{g}$.
- Viscous stresses negligible in the wake and on the bounding streamlines; $p - p_\infty \simeq 0$ on the whole outer boundary of the control volume (given). On the airfoil surface pressure and viscous stresses are **not** neglected: they make the drag.
- Wake profile at $x = x_0$ (symmetric about $y = 0$): $v_x = \frac{U}{2}\left(1 + \frac{|y|}{y_0}\right)$ for $|y| \leq y_0$, and $v_x = U$ outside.
- Data: $[\rho,\ U,\ y_0]$. Unknowns: $[y_{-\infty},\ D]$, with D the drag per unit span.

Symbol	Meaning	SI unit
U	uniform upstream velocity	m/s
y_0	half-width of the wake at $x = x_0$	m
$y_{-\infty}$	half-width of the stream tube far upstream	m
ρ	air density	kg/m^3
D	drag per unit span	N/m

Dimensional check of the unknown: $[\rho U^2 y_0] = (\mathrm{kg/m^3})(\mathrm{m^2/s^2})\,\mathrm{m} = \mathrm{N/m}$, a force per unit length, as required for a planar problem.

PHASE 2

Frames, Control Volume and Surfaces

- Inertial frame fixed to the airfoil and the tunnel; basis $(\bar{e}_x, \bar{e}_y, \bar{e}_z)$ with $\bar{e}_x$ along the stream. The control volume is fixed: $\bar{v}_c = 0$.
- Surfaces of Σ_c (outward normal $\bar{n}$): Σ_1, far upstream, $|y| < y_{-\infty}$, $\bar{n} = -\bar{e}_x$; Σ_2, the section $x = x_0$, $|y| < y_0$, $\bar{n} = +\bar{e}_x$; Σ_l, the two bounding streamlines ($\bar{v} \cdot \bar{n} = 0$); Σ_w, the airfoil surface ($\bar{n}$ pointing into the airfoil).
- Change of variable for the wake integrals: $s = 1 + y/y_0$, $dy = y_0\, ds$; $y \in [0, y_0]$ maps to $s \in [1, 2]$.

Force of the air on the airfoil (Notes Eq. 3.32), written on the wetted surface with the normal $\bar{n}$ of the control volume:

$$\boxed{\bar{F} = \int_{\Sigma_w} (p - p_\infty)\,\bar{n}\,d\sigma - \int_{\Sigma_w} \bar{\bar{\tau}}' \cdot \bar{n}\,d\sigma}$$

Hence the Σ_w terms of the momentum balance add up to $-\bar{F}$.

PHASE 3

Balances, Surface by Surface

`Mass (Notes Eq. 3.11)`

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\,(\bar{v} - \underbrace{\cancel{\bar{v}_c}}_{0})\cdot\bar{n}\,d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,\bar{v}\cdot\bar{n}\,d\sigma$
Σ_1	$\rho U\,(-1)\,2y_{-\infty} = -2\rho U y_{-\infty}$
Σ_2	$2\rho\int_0^{y_0} \frac{U}{2}\left(1+\frac{y}{y_0}\right)dy = \frac{3}{2}\rho U y_0$
Σ_l	0 (streamlines, $\bar{v}\cdot\bar{n}=0$)
Σ_w	0 (impermeable wall)

$$-2\rho U y_{-\infty} + \frac{3}{2}\rho U y_0 = 0 \implies \boxed{y_{-\infty} = \frac{3}{4}\,y_0} \qquad (1)$$

The Σ_2 entry in detail: the antiderivative of $1 + y/y_0$ is $y + y^2/(2y_0)$, and by Barrow's rule between $y=0$ and $y=y_0$, $\rho U\left[y + \frac{y^2}{2y_0}\right]_0^{y_0} = \rho U\left(y_0 + \frac{y_0}{2}\right) = \frac{3}{2}\rho U y_0$ (the factor 2 of the table counts both halves of the wake).

`Linear momentum (Notes Eq. 3.36, with` $p_a = p_\infty$`)`

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}}+\int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}]\,d\sigma=-\int_{\Sigma_c}(p-p_\infty)\,\bar{n}\,d\sigma+\int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma+\underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int(p-p_\infty)\,\bar{n}\,d\sigma$	$\int\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_1	$\rho U\,(-U)\,2y_{-\infty}\,\bar{e}_x=-2\rho U^2 y_{-\infty}\,\bar{e}_x$	0 ($p\simeq p_\infty$)	0 (uniform flow)
Σ_2	$2\rho\int_0^{y_0}v_x^2\,dy\;\bar{e}_x=\frac{7}{6}\rho U^2 y_0\,\bar{e}_x$	0 ($p\simeq p_\infty$)	0 (given)
Σ_l	0 ($\bar{v}\cdot\bar{n}=0$)	0 ($p\simeq p_\infty$)	0 (given)
Σ_w	0 **(impermeable)**	$-\int_{\Sigma_w}(p-p_\infty)\,\bar{n}\,d\sigma$	$\int_{\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed Σ_w terms: the force of the airfoil on the air, $-\bar{F}$. The Σ_1 entry uses (1): $-2\rho U^2 y_{-\infty}=-\frac{3}{2}\rho U^2 y_0$.

$$-\frac{3}{2}\rho U^2 y_0\,\bar{e}_x+\frac{7}{6}\rho U^2 y_0\,\bar{e}_x=-\bar{F}\implies\boxed{\bar{F}=D\,\bar{e}_x,\qquad D=\frac{1}{3}\,\rho\,U^2\,y_0}\quad(2)$$

The Σ_2 momentum entry in detail, with $s=1+y/y_0$, $dy=y_0\,ds$ and Barrow's rule between $s=1$ and $s=2$: $2\rho\frac{U^2}{4}\int_0^{y_0}\left(1+\frac{y}{y_0}\right)^2dy=\frac{\rho U^2 y_0}{2}\int_1^2 s^2\,ds=\frac{\rho U^2 y_0}{2}\left[\frac{s^3}{3}\right]_1^2=\frac{\rho U^2 y_0}{2}\cdot\frac{8-1}{3}=\frac{7}{6}\rho U^2 y_0$. The sum of the flux column is $\left(-\frac{9}{6}+\frac{7}{6}\right)\rho U^2 y_0=-\frac{1}{3}\rho U^2 y_0$.

For a wake symmetric about $y=0$ (a symmetric airfoil at zero incidence, as the symmetric profile implies) the y component of the momentum flux through Σ_2 cancels between the two halves, so $F_y=0$: no lift.

PHASE 4

Interpretation, Limits and Dimensional Check

- The wake carries less mass flux ($\frac{3}{2}\rho U y_0$) than the undisturbed stream through the same width ($2\rho U y_0$): the bounding streamlines are pushed outwards, from $y_{-\infty}=\frac{3}{4}y_0$ to y_0. The momentum deficit left in the wake is the drag.
- Limit $y_0\to 0$: no wake, $y_{-\infty}\to 0$ and $D\to 0$ (d'Alembert's paradox for a frictionless attached flow).

- Dimensions (SI): $[y_{-\infty}] = \mathrm{m}$; $[D] = [\rho U^2 y_0] = \mathrm{N/m}$.
- Numerical cross-check (midpoint quadrature of the defining integrals with 2×10^5 cells, $U = 1$ m/s, $\rho = 1.2\ \mathrm{kg/m^3}$, $y_0 = 0.35$ m): $y_{-\infty}/y_0 = 0.75000$ and $D = 0.140000$ N/m $= \rho U^2 y_0/3$.
- Agreement with the official solution: CL1.pdf obtains $y_{-\infty} = \frac{3}{4} y_0$ and $\bar{F}_{\text{airfoil}} = \frac{1}{3}\rho U^2 y_0\, \bar{e}_x$.

Extension (beyond the official solution)

With the momentum thickness $\theta = \int_{-y_0}^{y_0} \frac{v_x}{U}\left(1 - \frac{v_x}{U}\right) dy = 2y_0 \int_0^1 \frac{1-\sigma^2}{4}\, d\sigma = \frac{y_0}{3}$ ($\sigma = y/y_0$), the result reads $D = \rho U^2 \theta$, the classical momentum-deficit formula for the drag from a wake survey.

PROBLEM CL2

Bore over a Stagnant Water Layer

Original statement · CL2.pdf

Moving frame | Mass and momentum | Hydrostatic pressure | Froude number

OFFICIAL PROBLEM STATEMENT

The bore sketched in the figure of CL2.pdf moves under the action of gravity with constant velocity U over an initially stagnant water layer of thickness h_0, leaving behind a layer of thickness h_1 moving with velocity U_1. Assuming that in the air both viscous stresses and pressure variations with respect to the ambient value p_a are negligible, and so are viscous stresses on the bottom wall, find the values of U_1 and h_1 as a function of U, h_0 and g. As a preliminary computation, obtain the pressure distributions both upstream and downstream from the bore, verifying that they can be expressed in the form $p = p_a + \rho g(h_0 - z)$ and $p = p_a + \rho g(h_1 - z)$, respectively, where z represents the vertical distance to the bottom wall.

Source: CL2.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eqs. 3.10-3.11, integral momentum balance Eqs. 3.35-3.36.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Planar flow (per unit width) of an incompressible liquid of density ρ, with gravity $\bar{g} = -g\,\bar{e}_z$ (here gravity is **not** neglected: it drives the bore).
- The bore advances towards $-\bar{e}_x$ with constant velocity U over the stagnant layer h_0; behind it the layer h_1 moves with $-U_1\bar{e}_x$. Far from the front the velocity is uniform and horizontal and the free surface is flat at $p = p_a$.
- Viscous stresses negligible in the water sections and on the bottom wall (given); $\mu/(\rho U h) \ll 1$.
- Data: $[U,\ h_0,\ g,\ \rho]$. Unknowns: $[h_1,\ U_1]$ and the pressure distributions.

Symbol	Meaning	SI unit
U	velocity of the bore	m/s
U_1	velocity of the water behind the bore	m/s

Symbol	Meaning	SI unit
h_0, h_1	layer thickness ahead of and behind the bore	m
$F = U^2/(g h_0)$	Froude number of the bore	1

PHASE 2

Frames, Control Volume and Surfaces

- Laboratory frame L, with z measured from the bottom: unsteady flow, because the front moves.
- Frame of the bore B, moving with $-U\bar{e}_x$ relative to L. It is inertial (constant velocity), so the balances keep their form, and the flow is steady in it. Velocities: $\bar{v}_B = \bar{v}_L + U\bar{e}_x$, so the water ahead has $U\bar{e}_x$ and the water behind $(U - U_1)\bar{e}_x$.
- Control volume fixed in B ($\bar{v}_c = 0$). Surfaces: Σ_0, vertical section ahead ($\bar{n} = -\bar{e}_x$, height h_0); Σ_1, vertical section behind ($\bar{n} = +\bar{e}_x$, height h_1); Σ_b, bottom wall ($\bar{n} = -\bar{e}_z$); Σ_s, free surface (steady material surface in B, $p = p_a$).

PHASE 3

Balances, Surface by Surface

3.1 Preliminary: hydrostatic pressure in the uniform regions

Control volume: a column of width Δx in the uniform region ahead, from height z up to the free surface $z = h_0$. Surfaces: Σ_{bot} (at height z, $\bar{n} = -\bar{e}_z$), Σ_{top} (free surface, $\bar{n} = +\bar{e}_z$) and two vertical faces Σ_{side} ($\bar{n} = \mp\bar{e}_x$). Only the z component is needed.

> Momentum, z component, on the column (Notes Eq. 3.36)
>
> $$\underbrace{\frac{d}{dt}\int_{V_c} \rho v_z \, dV}_{0} + \int_{\Sigma_c} \rho v_z \, (\bar{v}\cdot\bar{n}) \, d\sigma = -\int_{\Sigma_c} p \, n_z \, d\sigma + \underbrace{\int_{\Sigma_c} (\bar{\bar{\tau}}'\cdot\bar{n})_z \, d\sigma}_{\text{negligible}} - \int_{V_c} \rho g \, dV$$
>
> Each surface integral, surface by surface (the vertical line drawn under each integral in class):
>
> | Surface | $\int \rho v_z \, (\bar{v}\cdot\bar{n}) \, d\sigma$ | $-\int p \, n_z \, d\sigma$ |
> |---|---|---|
> | Σ_{bot} | 0 ($v_z = 0$) | $+p(z)\,\Delta x$ |
> | Σ_{top} | 0 ($v_z = 0$) | $-p_a\,\Delta x$ |

Surface	$\int \rho v_z\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,n_z\,d\sigma$
Σ_{side}	0 ($v_z = 0$)	0 ($n_z = 0$)

Volume term: $-\int_{V_c} \rho g\,dV = -\rho g\,(h_0 - z)\,\Delta x$ (weight of the column).

$$0 = p(z)\,\Delta x - p_a\,\Delta x - \rho g\,(h_0 - z)\,\Delta x \implies \boxed{p = p_a + \rho g\,(h_0 - z)} \quad (1)$$

The same column taken behind the bore gives $p = p_a + \rho g\,(h_1 - z)$ (1').

3.2 Mass and momentum in the frame of the bore

`Mass (Notes Eq. 3.11), frame B`

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c} \rho\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\,(\bar{v} - \underbrace{\cancel{\bar{v}_c}}_{0})\cdot\bar{n}\,d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,\bar{v}\cdot\bar{n}\,d\sigma$
Σ_0	$\rho\,U\,(-1)\,h_0 = -\rho U h_0$
Σ_1	$\rho\,(U - U_1)\,h_1$
Σ_b	0 (wall)
Σ_s	0 (material surface)

$$\boxed{U\,h_0 = (U - U_1)\,h_1} \quad (2)$$

In the momentum balance p may be replaced by $p - p_a$ because $\int_{\Sigma_c} p_a \bar{n}\,d\sigma = 0$ on a closed surface. The x component is taken (gravity has no x component). On Σ_0 and Σ_1 the pressure is (1) and (1'); the antiderivative of $\rho g(h - z)$ is $\rho g(hz - z^2/2)$, and Barrow's rule between $z = 0$ and $z = h$ gives $\rho g h^2/2$.

Momentum, x component (Notes Eq. 3.36), frame B

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho v_x\,dV}}_{0\ (\text{steady in } B)} + \int_{\Sigma_c}\rho v_x\,(\bar{v}\cdot\bar{n})\,d\sigma = -\int_{\Sigma_c}(p-p_a)\,n_x\,d\sigma + \int_{\Sigma_c}(\bar{\bar{\tau}}'\cdot\bar{n})_x\,d\sigma + \underbrace{\cancel{\int_{V_c}\rho g_x\,dV}}_{g_x=0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho v_x\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int (p-p_a)\,n_x\,d\sigma$	$\int (\bar{\bar{\tau}}'\cdot\bar{n})_x\,d\sigma$
Σ_0	$\rho U\,(-U)\,h_0 = -\rho U^2 h_0$	$+\int_0^{h_0}\rho g(h_0-z)\,dz = \frac{\rho g h_0^2}{2}$	0
Σ_1	$\rho\,(U-U_1)^2\,h_1$	$-\int_0^{h_1}\rho g(h_1-z)\,dz = -\frac{\rho g h_1^2}{2}$	0
Σ_b	0	0 ($n_x=0$)	0 (given)
Σ_s	0	0 ($p=p_a$)	0 (air)

$$\boxed{-\rho U^2 h_0 + \rho\,(U-U_1)^2\,h_1 = \frac{\rho g}{2}\left(h_0^2 - h_1^2\right)} \qquad (3)$$

3.3 Solution for h_1/h_0 and U_1/U

From (2), $U - U_1 = U h_0/h_1$, so $(U-U_1)^2 h_1 = U^2 h_0^2/h_1$. Substituting into (3) and dividing by ρ:

$$U^2 h_0\left(\frac{h_0}{h_1} - 1\right) = \frac{g}{2}(h_0 - h_1)(h_0 + h_1) \implies U^2 h_0\,\frac{h_0 - h_1}{h_1} = \frac{g}{2}(h_0 - h_1)(h_0 + h_1).$$

For a bore $h_1 \neq h_0$, so $h_0 - h_1$ cancels. With $r = h_1/h_0$ and $F = U^2/(g h_0)$, dividing by $g h_0^2$ gives $F/r = (1+r)/2$:

$$r^2 + r - 2F = 0 \implies \boxed{\frac{h_1}{h_0} = \frac{1}{2}\left(\sqrt{1+8F} - 1\right)} \qquad (4)$$

(the negative root is rejected, $h_1 > 0$). From (2), $U_1/U = 1 - h_0/h_1 = (r-1)/r$; with $s = \sqrt{1+8F}$, $r = (s-1)/2$ and

$$\boxed{\frac{U_1}{U} = 1 - \frac{2}{s-1} = \frac{\sqrt{1+8F}-3}{\sqrt{1+8F}-1}}, \qquad F = \frac{U^2}{g h_0}. \tag{5}$$

Erratum in the official solution (reading of the handwriting)

CL2.pdf gives $h_1/h_0 = \frac{1}{2}(\sqrt{1+8F}-1)$, as in (4). For the velocity it gives, as legible in the scan, $U_1/U = \frac{3}{2}\left(\frac{1}{3}\sqrt{1+8F}-1\right) = (\sqrt{1+8F}-3)/2 = r-1$. This contradicts its own continuity equation (2), which requires $U_1/U = (r-1)/r$; the two coincide only for $r = 1$. Check: (2) gives $(U-U_1)h_1 = U h_0 > 0$, hence $U_1 < U$, whereas $r - 1 > 1$ for $F > 3$ (for $F = 4$, $r - 1 = 1.3723$ against $U_1/U = 0.5785$ from (5)). The computed value is kept.

PHASE 4

Interpretation, Limits and Dimensional Check

- A bore exists only for $F > 1$: $r > 1 \iff s > 3 \iff F > 1$. F compares the speed of the front with the shallow-water wave speed $\sqrt{g h_0}$.
- Weak bore, $F = 1 + \epsilon$: $r \approx 1 + \frac{2}{3}\epsilon$ and $U_1/U \approx \frac{2}{3}\epsilon \to 0$ (an infinitesimal gravity wave travelling at $\sqrt{g h_0}$).
- Strong bore, $F \gg 1$: $r \approx \sqrt{2F}$ and $U_1 \to U$.
- Dimensions (SI): both sides of (3) are forces per unit width, $[\rho U^2 h_0] = [\rho g h^2] = \mathrm{N/m}$; F, r and U_1/U are dimensionless.
- Numerical cross-check: $F = 4$, $h_0 = 0.5$ m, $g = 9.81$ m/s^2 give $U = 4.4294$ m/s, $h_1/h_0 = 2.3723$ ($h_1 = 1.1861$ m), $U_1/U = 0.57846$ ($U_1 = 2.5623$ m/s); the residuals of (2) and (3) are below 10^{-14} in SI units.

Extension (beyond the official solution)

Mechanical energy is not conserved across the bore, so Bernoulli's equation cannot be used between both sections. The head lost is the classical hydraulic-jump value $\Delta H = (h_1 - h_0)^3/(4 h_0 h_1)$, positive for $h_1 > h_0$.

PROBLEM CL3

Double-Elbow Mouthpiece: Exit Velocity, Force and Moment

Original statement · CL3.pdf

Control volume | Linear momentum | Angular momentum | Gauge pressure

OFFICIAL PROBLEM STATEMENT

The double-elbow mouth piece of exit cross section A_o shown in the figure of CL3.pdf is to be installed at the end of a circular pipe of cross section A_p to redirect the outflow of water into the open atmosphere. Once the mouth piece is installed, the uniform values U_p and p_p are measured for the flow velocity and the pressure at the end of the pipe (i.e., just at the entrance of the mouth piece). Find the value of the uniform velocity at the exit section, U_o, as well as the force exerted on the mouth piece and its corresponding moment with respect to the entrance section. (Handwritten addition: neglect the effect of gravity.)

Source: CL3.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral angular momentum balance Eqs. 3.37-3.38, force and moment on a body Eqs. 3.32-3.33.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, incompressible (water); neglect $\bar{g}$ (handwritten addition); uniform properties at inlet and outlet; viscous stresses negligible on both sections.
- Free jet at the exit: $p = p_a$ over Σ_o. The thin wall of the mouth piece has the atmosphere p_a on its outer face; the force asked for is the net force of the water inside plus the atmosphere outside, as in the official solution. Ignoring the atmosphere would add $p_a(A_p - A_o)\,\bar{e}_x$.
- Geometry (figure): S-shaped piece; the exit axis is a vertical distance H below the entrance axis; both the entrance and the exit flows are along $+\bar{e}_x$.
- Data: $[\rho,\ A_p,\ A_o,\ U_p,\ p_p,\ p_a,\ H]$. Unknowns: $[U_o,\ \bar{F}_{MP},\ \bar{M}_{MP}]$.

Symbol	Meaning	SI unit
A_p, A_o	entrance and exit areas	m^2
U_p, p_p	velocity and pressure at the entrance	m/s, Pa

Symbol	Meaning	SI unit
U_o	exit velocity	m/s
H	vertical offset between the axes	m
$\bar{F}_{MP}$, $\bar{M}_{MP}$	force and moment on the mouth piece	N, N·m

PHASE 2

Frames, Control Volume and Surfaces

- Inertial frame fixed to the pipe: $\bar{e}_x$ along the pipe axis (flow direction), $\bar{e}_y$ vertical upwards, $\bar{e}_z = \bar{e}_x \wedge \bar{e}_y$ towards the reader. Origin O at the centre of the entrance section (moments about O).
- Centre of the exit section: $\bar{x}_o = X\bar{e}_x - H\bar{e}_y$ relative to O (X drops out). Position within a circular section relative to its centre: $\bar{\eta}$, with $\int \bar{\eta}\, d\sigma = 0$.
- Control volume: the water inside the mouth piece, fixed ($\bar{v}_c = 0$). Surfaces: Σ_p (entrance, $\bar{n} = -\bar{e}_x$), Σ_o (exit, $\bar{n} = +\bar{e}_x$), Σ_w (inner wall).

Force and moment of the water and the atmosphere on the mouth piece, with $\bar{n}$ the outward normal of the control volume:

$$\boxed{\bar{F}_{MP} = \int_{\Sigma_w} (p - p_a)\, \bar{n}\, d\sigma - \int_{\Sigma_w} \bar{\bar{\tau}}' \cdot \bar{n}\, d\sigma}$$

$$\boxed{\bar{M}_{MP} = \int_{\Sigma_w} \bar{x} \wedge \left[(p - p_a)\, \bar{n} - \bar{\bar{\tau}}' \cdot \bar{n}\right] d\sigma}$$

PHASE 3

Balances, Surface by Surface

Mass (Notes Eq. 3.11)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho\,dV}}_{0\ \text{(steady)}}+\int_{\Sigma_c}\rho\,(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}\,d\sigma=0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,\bar{v}\cdot\bar{n}\,d\sigma$
Σ_p	$\rho\,U_p\bar{e}_x\cdot(-\bar{e}_x)\,A_p=-\rho U_pA_p$
Σ_o	$\rho\,U_o\bar{e}_x\cdot\bar{e}_x\,A_o=+\rho U_oA_o$
Σ_w	0 (wall)

$$\rho U_pA_p=\rho U_oA_o\implies\boxed{U_o=\frac{A_p}{A_o}\,U_p},\qquad\dot{m}\equiv\rho U_pA_p=\rho U_oA_o\quad(1)$$

Linear momentum (Notes Eq. 3.36)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}}+\int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}]\,d\sigma=-\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma+\int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma+\underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int\rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int(p-p_a)\,\bar{n}\,d\sigma$	$\int\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_p	$\rho U_p\bar{e}_x\,(-U_p)\,A_p=-\dot{m}\,U_p\bar{e}_x$	$-(p_p-p_a)(-\bar{e}_x)A_p=(p_p-p_a)A_p\bar{e}_x$	0 (uniform)
Σ_o	$\rho U_o\bar{e}_x\,(U_o)\,A_o=\dot{m}\,U_o\bar{e}_x$	0 (free jet, $p=p_a$)	0 (uniform)
Σ_w	0	$-\int_{\Sigma_w}(p-p_a)\,\bar{n}\,d\sigma$	$\int_{\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed Σ_w terms: $-\bar{F}_{MP}$.

$$\dot{m}\,(U_o - U_p)\,\bar{e}_x = (p_p - p_a)A_p\,\bar{e}_x - \bar{F}_{MP} \implies \boxed{\bar{F}_{MP} = \left[(p_p - p_a)A_p + \rho U_p\,(U_p - U_o)\,A_p\right]\bar{e}_x} \quad (2)$$

No entry of the table has a component along $\bar{e}_y$ or $\bar{e}_z$, so $F_{MP,y} = F_{MP,z} = 0$.

Angular momentum about O (Notes Eq. 3.38)

$$\underbrace{\frac{d}{dt}\int_{V_c} \cancel{\rho\,\bar{x}\wedge\bar{v}\,dV}}_{0} + \int_{\Sigma_c} \rho\,(\bar{x}\wedge\bar{v})(\bar{v}\cdot\bar{n})\,d\sigma = -\int_{\Sigma_c} \bar{x}\wedge(p-p_a)\bar{n}\,d\sigma + \int_{\Sigma_c} \bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma + \underbrace{\int_{V_c} \cancel{\rho\,\bar{x}\wedge\bar{g}\,dV}}_{\text{neglect } \bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,(\bar{x}\wedge\bar{v})(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int \bar{x}\wedge(p-p_a)\bar{n}\,d\sigma$	$\int \bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$
Σ_p	$-\rho U_p^2\left(\int \bar{\eta}\,d\sigma\right)\wedge\bar{e}_x = 0$	$(p_p - p_a)\left(\int \bar{\eta}\,d\sigma\right)\wedge\bar{e}_x = 0$	0
Σ_o	$\rho U_o^2 A_o\,(\bar{x}_o\wedge\bar{e}_x) = \rho U_o^2 H A_o\,\bar{e}_z$	$0\ (p = p_a)$	0
Σ_w	0	$-\int_{\Sigma_w} \bar{x}\wedge(p-p_a)\bar{n}\,d\sigma$	$\int_{\Sigma_w} \bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$

Boxed Σ_w terms: $-\bar{M}_{MP}$. In the Σ_o entry, $\bar{x} = \bar{x}_o + \bar{\eta}$ and $\bar{x}_o\wedge\bar{e}_x = \left(X\bar{e}_x - H\bar{e}_y\right)\wedge\bar{e}_x = -H\,\bar{e}_y\wedge\bar{e}_x = +H\,\bar{e}_z$.

$$\rho U_o^2 H A_o\,\bar{e}_z = -\bar{M}_{MP} \implies \boxed{\bar{M}_{MP} = -\rho U_o^2 H A_o\,\bar{e}_z = -\rho\,\frac{A_p^2}{A_o}\,U_p^2\,H\,\bar{e}_z} \quad (3)$$

PHASE 4

Interpretation, Limits and Dimensional Check

- $A_o < A_p$ gives $U_o > U_p$: the piece accelerates the water, and the wall supplies the momentum change.

- The moment exists only because the exit axis is offset by H: for $H \to 0$, $\bar{M}_{MP} \to 0$. With the exit below the entrance, the moment is clockwise as seen in the figure ($\bar{e}_z$ towards the reader).
- Limit $A_o \to A_p$: $U_o = U_p$ and $\bar{F}_{MP} \to (p_p - p_a) A_p \bar{e}_x$, the pure pressure force.
- Dimensions (SI): $[\rho U_p A_p (U_p - U_o)] = \mathrm{N}$; $[\rho U_o^2 H A_o] = \mathrm{N \cdot m}$.
- Numerical illustration (hypothetical data): $\rho = 1000$ kg/m^3, $A_p = 0.02$ m^2, $A_o = 0.008$ m^2, $U_p = 3$ m/s, $p_p = 2.5 \times 10^5$ Pa, $p_a = 101325$ Pa, $H = 1.5$ m give $U_o = 7.5$ m/s, $F_{MP,x} = 2703.5$ N and $M_{MP,z} = -675$ N·m.
- Agreement with the official solution: CL3.pdf obtains (1), (2) and (3).

Extension (beyond the official solution)

If the flow inside were lossless (not stated), Bernoulli's equation $p_p - p_a = \frac{1}{2}\rho(U_o^2 - U_p^2)$ would give $F_{MP,x} = \frac{1}{2}\rho A_p (U_o - U_p)^2 \geq 0$. With losses, the measured p_p must be used, as in (2).

PROBLEM CL4

Cascade of Aerofoils: Downstream State

Original statement · CL4.pdf

Periodicity | Mass, momentum, energy | Perfect gas | Cascade

OFFICIAL PROBLEM STATEMENT

A planar uniform stream of air with pressure p_1, density ρ_1 and velocity $\bar{v}_1 = v_{1x}\bar{e}_x$ flows past an array of aerofoils with vertical separation L, exerting on each one a force (per unit length) given by $\bar{F} = F_x\bar{e}_x + F_y\bar{e}_y$. Assuming that the aerofoils are thermally insulated and neglecting the effect of gravity, write the equations that determine the values of the pressure p_2, density ρ_2 and velocity $\bar{v}_2 = v_{2x}\bar{e}_x + v_{2y}\bar{e}_y$ for the uniform stream found sufficiently far downstream.

Source: CL4.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral total energy balance Eq. 3.46, force on a body Eq. 3.32.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady planar flow through an infinite array of identical aerofoils (pitch L), so the flow is periodic in y; uniform far upstream (1) and far downstream (2).
- Neglect $\bar{g}$; aerofoils thermally insulated; compressible fluid. Viscous stresses and heat conduction negligible in the uniform sections.
- Perfect gas: $h = e + \frac{p}{\rho} = c_pT = \frac{c_p}{R_g}\frac{p}{\rho} = \frac{\gamma}{\gamma-1}\frac{p}{\rho}$ (with $p = \rho R_g T$ and $\gamma = c_p/c_v$), as in the class solution.
- Data: $[p_1,\ \rho_1,\ v_{1x},\ L,\ F_x,\ F_y,\ \gamma]$. Unknowns: $[p_2,\ \rho_2,\ v_{2x},\ v_{2y}]$: four equations are needed.

Symbol	Meaning	SI unit
p_1, ρ_1, v_{1x}	upstream state	Pa, kg/m^3, m/s
p_2, ρ_2, v_{2x}, v_{2y}	downstream state	Pa, kg/m^3, m/s
L	pitch of the array	m

Symbol	Meaning	SI unit
F_x, F_y	force on each aerofoil per unit length	N/m

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the array; $\bar{e}_x$ along the incoming stream, $\bar{e}_y$ along the array. Fixed control volume, $\bar{v}_c = 0$, one period high.
- Surfaces: Σ_1 (far upstream, height L, $\bar{n} = -\bar{e}_x$), Σ_2 (far downstream, height L, $\bar{n} = +\bar{e}_x$), Σ_{l1} and Σ_{l2} (upper and lower stream surfaces, $\bar{v} \cdot \bar{n} = 0$), Σ_w (aerofoil surface, $\bar{v} = 0$, insulated).
- Periodicity: Σ_{l2} is Σ_{l1} shifted by $-L\bar{e}_y$ with opposite normals, so the integrals of any periodic quantity over Σ_{l1} and Σ_{l2} cancel each other out.

Force on each aerofoil (per unit length), with $\bar{n}$ the outward normal of the control volume:

$$\boxed{\bar{F} = F_x\bar{e}_x + F_y\bar{e}_y = \int_{\Sigma_w} p\,\bar{n}\,d\sigma - \int_{\Sigma_w} \bar{\bar{\tau}}' \cdot \bar{n}\,d\sigma}$$

PHASE 3

Balances, Surface by Surface

Mass (Notes Eq. 3.11)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\,(\bar{v} - \underbrace{\bar{v}_c}_{0}) \cdot \bar{n}\,d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,\bar{v} \cdot \bar{n}\,d\sigma$
Σ_1	$\rho_1(-v_{1x})\,L$
Σ_2	$\rho_2\,v_{2x}\,L$

Surface	$\int \rho\,\bar{v}\cdot\bar{n}\,d\sigma$
$\Sigma_{l1},\ \Sigma_{l2}$	0 (stream surfaces)
Σ_w	0 (wall)

$$\boxed{\rho_1 v_{1x} = \rho_2 v_{2x}} \qquad (1)$$

`Linear momentum (Notes Eq. 3.36)`

$$\underbrace{\frac{d}{dt}\int_{V_c}\rho\bar{v}\,dV}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c} p\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\int_{V_c}\rho\bar{g}\,dV}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_1	$\rho_1 v_{1x}\bar{e}_x(-v_{1x})L$	$+p_1 L\,\bar{e}_x$	0
Σ_2	$\rho_2(v_{2x}\bar{e}_x + v_{2y}\bar{e}_y)\,v_{2x}L$	$-p_2 L\,\bar{e}_x$	0
$\Sigma_{l1}+\Sigma_{l2}$	0	0 (cancel each other out)	0 (cancel each other out)
Σ_w	0	$-\int_{\Sigma_w} p\,\bar{n}\,d\sigma$	$\int_{\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed Σ_w terms: $-\bar{F} = -F_x\bar{e}_x - F_y\bar{e}_y$. With (1), $\rho_2 v_{2x}L = \rho_1 v_{1x}L$ in the Σ_2 flux.

$$\boxed{F_x = \rho_1 v_{1x} L\,(v_{1x}-v_{2x}) + (p_1-p_2)\,L} \quad (2) \qquad \boxed{F_y = -\rho_1 v_{1x} L\,v_{2y}} \quad (3)$$

`Total energy (Notes Eq. 3.46)`

$$\underbrace{\frac{d}{dt}\int_{V_c}\rho\left(e+\frac{v^2}{2}\right)dV}_{0} + \int_{\Sigma_c}\rho\left(e+\frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma = -\int_{\Sigma_c} p\,\bar{v}\cdot\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma + \underbrace{\int_{V_c}\rho\bar{g}\cdot\bar{v}\,dV}_{\text{neglect}} - \int_{\Sigma_c}\bar{q}\cdot\bar{n}\,d\sigma + \underbrace{\int_{V_c}(Q_c+Q_r)\,dV}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e+\frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	$-\int \bar{q}\cdot\bar{n}\,d\sigma$
Σ_1	$\rho_1\left(e_1+\frac{v_{1x}^2}{2}\right)(-v_{1x})L$	$+p_1 v_{1x} L$	0	0
Σ_2	$\rho_2\left(e_2+\frac{v_{2x}^2+v_{2y}^2}{2}\right)v_{2x}L$	$-p_2 v_{2x} L$	0	0
$\Sigma_{l1}+\Sigma_{l2}$	0	0	0 (cancel each other out)	0 (cancel each other out)
Σ_w	0	0 ($\bar{v}=0$)	0 ($\bar{v}=0$)	0 (insulated)

Moving the pressure work to the left turns e into $h=e+p/\rho$; dividing by $\rho_1 v_{1x} L = \rho_2 v_{2x} L$ and using $h=\frac{\gamma}{\gamma-1}\frac{p}{\rho}$:

$$\boxed{\frac{\gamma}{\gamma-1}\left(\frac{p_2}{\rho_2}-\frac{p_1}{\rho_1}\right)+\frac{v_{2x}^2+v_{2y}^2}{2}-\frac{v_{1x}^2}{2}=0} \tag{4}$$

(1) to (4) are four equations for the four unknowns p_2, ρ_2, v_{2x}, v_{2y}, which answers the question.

PHASE 4

Interpretation, Limits and Dimensional Check

- (3): the transverse force on an aerofoil equals minus the y momentum given to the stream; the flow is deflected opposite to the force. (2) splits the streamwise force into a pressure drop and a momentum change.
- $\bar{F}=0$: (1) to (4) are satisfied by state 2 = state 1. Incompressible limit ($\rho_2=\rho_1$): $v_{2x}=v_{1x}$ and $p_1-p_2=F_x/L$.
- Dimensions (SI): $[F_x/L]=\mathrm{Pa}=[\rho_1 v_{1x}^2]$; $[\rho_1 v_{1x} L v_{2y}]=\mathrm{N/m}$; $[\gamma p/((\gamma-1)\rho)]=\mathrm{m^2/s^2}=[v^2]$.
- Agreement with the official solution and the class solution: the same four equations (1) to (4).

Extension (beyond the official solution): closed solution

With $m = \rho_1 v_{1x}$: $\rho_2 = m/v_{2x}$, $v_{2y} = -F_y/(mL)$ and $p_2 = P - m\,v_{2x}$ with $P = p_1 + mv_{1x} - F_x/L$. Substituting into (4), with $E = \frac{\gamma}{\gamma-1}\frac{p_1}{\rho_1} + \frac{v_{1x}^2}{2}$ and the coefficient $\frac{1}{2} - \frac{\gamma}{\gamma-1} = -\frac{\gamma+1}{2(\gamma-1)}$ of v_{2x}^2: $(\gamma+1)v_{2x}^2 - 2\gamma\frac{P}{m}v_{2x} + 2(\gamma-1)\left(E - \frac{v_{2y}^2}{2}\right) = 0$. For $\bar{F} = 0$ one root is $v_{2x} = v_{1x}$; the root continuous with it is the physical one. Check: $p_1 = 101325$ Pa, $\rho_1 = 1.2$ kg/m^3, $v_{1x} = 100$ m/s, $L = 0.2$ m, $F_x = 30$ N/m, $F_y = -80$ N/m, $\gamma = 1.4$ give $v_{2x} = 100.16$ m/s, $v_{2y} = 3.3333$ m/s, $\rho_2 = 1.19808$ kg/m^3, $p_2 = 101155.8$ Pa, residuals of (1) to (4) below 10^{-9}; the other root gives $p_2 = -6843.3$ Pa < 0 and is discarded.

PROBLEM CL5

Turbojet on a Testbench: Exit State

Original statement · CL5.pdf

Moving boundaries | Mass, momentum, energy | Heat and work | Thrust

OFFICIAL PROBLEM STATEMENT

The entrance and exit sections of the turboreactor sketched in the figure of CL5.pdf have areas A_1 and A_2. Besides the outer fixed frame, the machine has mobile parts in its interior that extract an amount of work W per unit time from the fluid stream. The combined effects of chemical reaction in the combustion chamber and heat transfer through the walls give a heat transfer per unit time Q to the fluid stream. The axial force on the turboreactor measured on the testbench is $\bar{F} = -F\,\bar{e}_x$. For given values of W, Q and F and assuming that at the entrance and exit sections the flow properties are uniform, obtain the values of the density ρ_2, pressure p_2 and velocity v_2 in terms of their values p_1, ρ_1 and v_1 at the entrance. (Handwritten addition: no gravity.)

Source: CL5.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eq. 3.10, integral momentum balance Eq. 3.35, integral total energy balance Eq. 3.46, force on a body Eq. 3.32.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Compressible flow; neglect gravity; uniform properties at inlet and outlet; steady on average (the rotating parts make the flow periodic inside, but there is no net accumulation).
- Perfect gas, $h = \frac{\gamma}{\gamma-1}\frac{p}{\rho}$, as in the class solution. Fuel mass flow neglected (not given), so the same mass flow crosses both sections.
- $\dot{W}$: work per unit time extracted from the fluid by the mobile parts; $\dot{Q}$: heat per unit time transferred to the fluid; $\bar{F} = -F\bar{e}_x$: force on the device measured on the testbench (fluid inside plus ambient air outside, at p_a).
- Data: $[\rho_1,\ p_1,\ \bar{v}_1 = v_1\bar{e}_x,\ A_1,\ A_2,\ p_a,\ F,\ \dot{W},\ \dot{Q},\ \gamma]$. Unknowns: $[\rho_2,\ p_2,\ \bar{v}_2 = v_2\bar{e}_x]$.

Symbol	Meaning	SI unit
A_1, A_2	entrance and exit areas	m^2

Symbol	Meaning	SI unit
p_1, ρ_1, v_1	entrance state	Pa, kg/m^3, m/s
p_2, ρ_2, v_2	exit state	Pa, kg/m^3, m/s
F	thrust measured on the bench	N
$\dot{W}, \dot{Q}$	power extracted and heat added	W
$\dot{m} = \rho_1 v_1 A_1$	mass flow	kg/s

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the testbench; $\bar{e}_x$ along the engine axis in the flow direction.
- Surfaces: Σ_1 (entrance, $\bar{n} = -\bar{e}_x$), Σ_2 (exit, $\bar{n} = +\bar{e}_x$), Σ_w (fixed walls, $\bar{v} = \bar{v}_c = 0$) and Σ_m (surface of the mobile parts, moving with them: $\bar{v} = \bar{v}_c \neq 0$). On Σ_w and Σ_m, $(\bar{v} - \bar{v}_c) \cdot \bar{n} = 0$.
- $\bar{n}$ is the outward normal of the control volume; $\bar{n}_{\text{in}} = -\bar{n}$ points into the fluid.

Definitions written on the surfaces, as in class:

$$\boxed{\bar{F} = \int_{\Sigma_m, \Sigma_w} (p - p_a)\, \bar{n}\, d\sigma - \int_{\Sigma_m, \Sigma_w} \bar{\bar{\tau}}' \cdot \bar{n}\, d\sigma = -F\, \bar{e}_x}$$

$$\boxed{\dot{Q} = \int_{\Sigma_w, \Sigma_m} \bar{q} \cdot \bar{n}_{\text{in}}\, d\sigma + \int_{V_c} Q_R\, dV}$$

$$\boxed{\dot{W} = -\int_{\Sigma_m} p\, \bar{n}_{\text{in}} \cdot \bar{v}\, d\sigma + \int_{\Sigma_m} (\bar{\bar{\tau}}' \cdot \bar{n}_{\text{in}}) \cdot \bar{v}\, d\sigma}$$

Only Σ_m enters $\dot{W}$: Σ_w is a fixed surface and does no work. $\dot{W}$ is the power of the gas on the moving solid, so the power done on the gas there is $-\dot{W}$.

PHASE 3

Balances, Surface by Surface

Mass (Notes Eq. 3.10)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho \, dV}}_{0 \text{ (on average)}} + \int_{\Sigma_c} \rho \left(\bar{v} - \bar{v}_c\right) \cdot \bar{n} \, d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho \left(\bar{v} - \bar{v}_c\right) \cdot \bar{n} \, d\sigma$
Σ_1	$-\rho_1 v_1 A_1$
Σ_2	$+\rho_2 v_2 A_2$
Σ_m	$0 \; (\bar{v} = \bar{v}_c)$
Σ_w	0

$$\boxed{\rho_1 v_1 A_1 = \rho_2 v_2 A_2 = \dot{m}} \qquad (1)$$

Linear momentum (Notes Eq. 3.35)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho \bar{v} \, dV}}_{0} + \int_{\Sigma_c} \rho \bar{v} \left[\left(\bar{v} - \bar{v}_c\right) \cdot \bar{n}\right] d\sigma = -\int_{\Sigma_c} \left(p - p_a\right) \bar{n} \, d\sigma + \int_{\Sigma_c} \bar{\bar{\tau}}' \cdot \bar{n} \, d\sigma + \underbrace{\cancel{\int_{V_c} \rho \bar{g} \, dV}}_{\text{neglect}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho \bar{v} \left[\left(\bar{v} - \bar{v}_c\right) \cdot \bar{n}\right] d\sigma$	$-\int \left(p - p_a\right) \bar{n} \, d\sigma$	$\int \bar{\bar{\tau}}' \cdot \bar{n} \, d\sigma$
Σ_1	$\rho_1 v_1 \bar{e}_x (-v_1) A_1 = -\dot{m} v_1 \bar{e}_x$	$\left(p_1 - p_a\right) A_1 \bar{e}_x$	0
Σ_2	$\rho_2 v_2 \bar{e}_x (v_2) A_2 = \dot{m} v_2 \bar{e}_x$	$-\left(p_2 - p_a\right) A_2 \bar{e}_x$	0
$\Sigma_m,\ \Sigma_w$	0	$-\int_{\Sigma_m, \Sigma_w} \left(p - p_a\right) \bar{n} \, d\sigma$	$\int_{\Sigma_m, \Sigma_w} \bar{\bar{\tau}}' \cdot \bar{n} \, d\sigma$

Boxed Σ_m, Σ_w terms: $-\bar{F} = F\bar{e}_x$ (uniform velocity vectors give no viscous stress on Σ_1, Σ_2).

$$\boxed{\dot{m}\,(v_2 - v_1) = (p_1 - p_a)A_1 - (p_2 - p_a)A_2 + F} \tag{2}$$

Total energy (Notes Eq. 3.46)

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho\left(e + \frac{v^2}{2}\right) dV}_{0} + \int_{\Sigma_c} \rho\left(e + \frac{v^2}{2}\right)[(\bar{v} - \bar{v}_c)\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c} p\,\bar{v}\cdot\bar{n}\,d\sigma + \int_{\Sigma_c} (\bar{\bar{\tau}}'\cdot\bar{n})\cdot\bar{v}\,d\sigma + \underbrace{\int_{V_c} \rho\bar{g}\cdot\bar{v}\,dV}_{\text{neglect}} - \int_{\Sigma_c} \bar{q}\cdot\bar{n}\,d\sigma + \int_{V_c} Q_R\,dV$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e + \frac{v^2}{2}\right)[(\bar{v} - \bar{v}_c)\cdot\bar{n}]\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int (\bar{\bar{\tau}}'\cdot\bar{n})\cdot\bar{v}\,d\sigma$	$-\int \bar{q}\cdot\bar{n}\,d\sigma$
Σ_1	$-\dot{m}\left(e_1 + \frac{v_1^2}{2}\right)$	$+p_1 v_1 A_1$	0	0 ($\nabla T = 0$)
Σ_2	$+\dot{m}\left(e_2 + \frac{v_2^2}{2}\right)$	$-p_2 v_2 A_2$	0	0 ($\nabla T = 0$)
Σ_m	0	$-\int_{\Sigma_m} p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int_{\Sigma_m} (\bar{\bar{\tau}}'\cdot\bar{n})\cdot\bar{v}\,d\sigma$	$-\int_{\Sigma_m} \bar{q}\cdot\bar{n}\,d\sigma$
Σ_w	0	0 ($\bar{v} = 0$)	0 ($\bar{v} = 0$)	$-\int_{\Sigma_w} \bar{q}\cdot\bar{n}\,d\sigma$

Boxed terms: the pressure and viscous work on Σ_m add up to $-\dot{W}$; the conduction through Σ_m, Σ_w plus the volume term $\int_{V_c} Q_R\,dV$ add up to $\dot{Q}$. Moving the pressure work on Σ_1, Σ_2 to the left gives $h = e + p/\rho$; with (1) and $h = \frac{\gamma}{\gamma-1}\frac{p}{\rho}$:

$$\boxed{\dot{m}\left[\left(\frac{\gamma}{\gamma-1}\frac{p_2}{\rho_2} + \frac{v_2^2}{2}\right) - \left(\frac{\gamma}{\gamma-1}\frac{p_1}{\rho_1} + \frac{v_1^2}{2}\right)\right] = -\dot{W} + \dot{Q}} \tag{3}$$

(1), (2) and (3) determine ρ_2, p_2 and v_2.

PHASE 4

Interpretation, Limits and Dimensional Check

- (2) is the thrust balance: thrust = change of momentum flux + pressure imbalance on the sections. For $p_1 = p_2 = p_a$, $F = \dot{m}(v_2 - v_1)$.
- (3): the stagnation enthalpy of the stream rises by the heat added and falls by the work extracted, per unit mass flow.
- Dimensions (SI): $[\dot{m}(v_2 - v_1)] = \mathrm{N}$; $[(\dot{Q} - \dot{W})/\dot{m}] = \mathrm{J/kg} = \mathrm{m^2/s^2} = [h] = [v^2]$.
- Numerical illustration (hypothetical data): $A_1 = 0.5$ m^2, $A_2 = 0.3$ m^2, $v_1 = 150$ m/s, $p_1 = 0.9 \times 10^5$ Pa, $\rho_1 = 1.1$ kg/m^3 ($M_1 = 0.443$), $p_a = 10^5$ Pa, $F = 30$ kN, $\dot{W} = 2$ MW, $\dot{Q} = 30$ MW, $\gamma = 1.4$: $\dot{m} = 82.5$ kg/s and the two roots of the extension below, $v_2 = 355.515$ m/s, $p_2 = 126817$ Pa, $\rho_2 = 0.77353$ kg/m^3 ($M_2 = 0.742$), and $v_2 = 597.263$ m/s, $p_2 = 60336$ Pa, $\rho_2 = 0.46043$ kg/m^3 ($M_2 = 1.39$); residuals of (1) to (3) below 10^{-6} in SI units.
- Agreement with the official and class solutions: the same equations (1), (2), (3) (CL5.pdf writes (2) as $F = \rho_1 v_1 A_1(v_2 - v_1) - (p_1 - p_a)A_1 + (p_2 - p_a)A_2$).

Extension (beyond the official solution): explicit solution

With $K = p_a A_2 + F + (p_1 - p_a)A_1 + \dot{m}v_1$, (2) gives $p_2 A_2 = K - \dot{m}v_2$ and, with (1), $p_2/\rho_2 = (K - \dot{m}v_2)v_2/\dot{m}$. Substituting into (3) and using $\frac{1}{2} - \frac{\gamma}{\gamma-1} = -\frac{\gamma+1}{2(\gamma-1)}$: $(\gamma+1)v_2^2 - 2\gamma\frac{K}{\dot{m}}v_2 + 2(\gamma-1)H_0 = 0$, with $H_0 = \frac{\gamma}{\gamma-1}\frac{p_1}{\rho_1} + \frac{v_1^2}{2} + \frac{\dot{Q}-\dot{W}}{\dot{m}}$. The two roots are a subsonic and a supersonic exit state; for a subsonic entrance with heat addition the physical one is the subsonic (smaller) root, since reaching the other would require passing through sonic conditions inside the engine, which this overall balance cannot describe. For $F = 0$, $\dot{Q} = \dot{W} = 0$ and $A_1 = A_2$ one root is $v_2 = v_1$ and the product of the roots is $2(\gamma-1)H_0/(\gamma+1)$, the Prandtl relation of a normal shock.

PROBLEM CL6

Elbow Discharging a Vertical Jet: Velocity, Force and Heat

Original statement · CL6.pdf

Elbow | Linear momentum | Energy | Heat through the walls

OFFICIAL PROBLEM STATEMENT

A liquid of density ρ and specific heat c flows through the elbow-shaped pipe shown in CL6.pdf, discharging to the atmosphere as a vertical jet through an outlet section of surface area A_o, different from the value A_i at the inlet. Gravity is assumed to have a negligible effect on the fluid motion. Besides, uniform flow properties are assumed at the inlet and outlet sections, where readings from a thermometer give different values T_i and T_o for the temperature. The velocity at the outlet section is measured to be v_o and a manometer is used to determine the value p_i for the pressure at the inlet section. In terms of all of these quantities, determine:

- The value of the velocity at the inlet v_i.
- The two components of the force (F_x, F_z) exerted on the pipe.
- The amount of heat Q transferred per unit time to the liquid by heat conduction through the pipe walls.

Source: CL6.pdf (page 1), also solved in class (class notes, CL6). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral total energy balance Eq. 3.46.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, incompressible liquid, $e = cT$; neglect $\bar{g}$; uniform properties at inlet and outlet (no viscous stresses or conduction across them).
- Free jet: $p = p_a$ over the outlet section. The outer face of the pipe is in contact with the ambient air at p_a, so the force on the pipe is computed with gauge pressure $p - p_a$.
- Geometry (figure): the liquid enters horizontally along $+\bar{e}_x$ and leaves as a vertical jet along $-\bar{e}_z$ ($\bar{e}_z$ upwards).
- Data: $[\rho,\ c,\ A_i,\ A_o,\ v_o,\ p_i,\ p_a,\ T_i,\ T_o]$. Unknowns: $[v_i,\ F_x,\ F_z,\ \dot{Q}]$.

Symbol	Meaning	SI unit
A_i, A_o	inlet and outlet areas	m^2
v_i, v_o	inlet and outlet velocities	m/s
p_i, p_a	inlet and ambient pressures	Pa
T_i, T_o	inlet and outlet temperatures	K
c	specific heat of the liquid	J/(kg·K)
$\dot{Q}$	heat per unit time into the liquid	W

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the pipe, $\bar{e}_x$ horizontal (inlet flow direction), $\bar{e}_z$ vertical upwards. Fixed control volume: the liquid inside the pipe, $\bar{v}_c = 0$.
- Surfaces: Σ_i (inlet, $\bar{n} = -\bar{e}_x$, $\bar{v} = v_i \bar{e}_x$), Σ_o (outlet, $\bar{n} = -\bar{e}_z$, $\bar{v} = -v_o \bar{e}_z$), Σ_w (pipe wall, $\bar{v} = 0$).

Force of the liquid and the ambient air on the pipe, and heat through the walls, written on Σ_w:

$$\boxed{\bar{F} = F_x \bar{e}_x + F_z \bar{e}_z = \int_{\Sigma_w} (p - p_a)\, \bar{n}\, d\sigma - \int_{\Sigma_w} \bar{\bar{\tau}}' \cdot \bar{n}\, d\sigma}$$

$$\boxed{\dot{Q} = -\int_{\Sigma_w} \bar{q} \cdot \bar{n}\, d\sigma}$$

PHASE 3

Balances, Surface by Surface

Mass (Notes Eq. 3.11)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\,(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}\,d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,\bar{v}\cdot\bar{n}\,d\sigma$
Σ_i	$\rho\,v_i\bar{e}_x\cdot(-\bar{e}_x)A_i = -\rho v_i A_i$
Σ_o	$\rho\,(-v_o\bar{e}_z)\cdot(-\bar{e}_z)A_o = +\rho v_o A_o$
Σ_w	0

$$\boxed{v_i = \frac{A_o}{A_i}\,v_o}, \qquad \dot{m} = \rho v_i A_i = \rho v_o A_o \tag{1}$$

Linear momentum (Notes Eq. 3.36)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int(p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_i	$\rho v_i\bar{e}_x\,(-v_i)A_i = -\rho v_i^2 A_i\,\bar{e}_x$	$-(p_i-p_a)(-\bar{e}_x)A_i = (p_i-p_a)A_i\,\bar{e}_x$	0
Σ_o	$\rho(-v_o\bar{e}_z)\,(v_o)A_o = -\rho v_o^2 A_o\,\bar{e}_z$	0 (free jet)	0
Σ_w	0	$-\int_{\Sigma_w}(p-p_a)\,\bar{n}\,d\sigma$	$\int_{\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed Σ_w terms: $-\bar{F}$.

$$-\rho v_i^2 A_i\,\bar{e}_x - \rho v_o^2 A_o\,\bar{e}_z = (p_i - p_a)A_i\,\bar{e}_x - \bar{F} \implies \boxed{F_x = \left[\rho v_i^2 + (p_i - p_a)\right] A_i, \qquad F_z = \rho v_o^2 A_o} \quad (2)$$

`Total energy (Notes Eq. 3.46)`

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho\left(e + \frac{v^2}{2}\right) dV}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\left(e + \frac{v^2}{2}\right)\left[(\bar{v} - \underbrace{\bar{v}_c}_{0})\cdot\bar{n}\right] d\sigma = -\int_{\Sigma_c} p\,\bar{v}\cdot\bar{n}\,d\sigma + \int_{\Sigma_c} \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma + \underbrace{\int_{V_c} \rho\bar{g}\cdot\bar{v}\,dV}_{\text{neglect } \bar{g}} - \int_{\Sigma_c} \bar{q}\cdot\bar{n}\,d\sigma + \underbrace{\int_{V_c} Q_H\,dV}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e + \frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	$-\int \bar{q}\cdot\bar{n}\,d\sigma$
Σ_i	$-\dot{m}\left(cT_i + \frac{v_i^2}{2}\right)$	$+p_i v_i A_i$	0	0 (uniform T)
Σ_o	$+\dot{m}\left(cT_o + \frac{v_o^2}{2}\right)$	$-p_a v_o A_o$	0	0 (uniform T)
Σ_w	0	0 ($\bar{v} = 0$)	0 ($\bar{v} = 0$)	$-\int_{\Sigma_w} \bar{q}\cdot\bar{n}\,d\sigma$

Boxed Σ_w term: $\dot{Q}$. Using (1), $p_i v_i A_i - p_a v_o A_o = (p_i - p_a)\,v_o A_o$.

$$\dot{m}\left[c(T_o - T_i) + \frac{v_o^2 - v_i^2}{2}\right] = (p_i - p_a)\,v_o A_o + \dot{Q} \implies \boxed{\dot{Q} = \rho v_o A_o\left[c(T_o - T_i) + \frac{v_o^2 - v_i^2}{2}\right] + (p_a - p_i)\,v_o A_o} \quad (3)$$

PHASE 4

Interpretation, Limits and Dimensional Check

- $F_x > 0$: the pipe is pushed along the incoming flow by the momentum flux and the gauge pressure at the inlet. $F_z > 0$: the jet leaves downwards, so the pipe is pushed upwards.
- (3) splits the heat into the rise of internal energy, the change of kinetic energy and the flow work. If no heat were exchanged and there were no losses, (3) with $\dot{Q} = 0$ and $T_o = T_i$ would reduce to Bernoulli's equation $p_i + \frac{1}{2}\rho v_i^2 = p_a + \frac{1}{2}\rho v_o^2$.

- Dimensions (SI): $[\rho v_i^2 A_i] = [(p_i - p_a) A_i] = \mathrm{N}$; $[\rho v_o A_o \, c \, \Delta T] = (\mathrm{kg/s})(\mathrm{J/(kg\,K)})\,\mathrm{K} = \mathrm{W}$; $[(p_a - p_i) v_o A_o] = \mathrm{Pa \cdot m^3/s} = \mathrm{W}$.
- Numerical illustration (hypothetical data): water, $c = 4186$ J/(kg K), $A_i = 0.01$ m^2, $A_o = 0.004$ m^2, $v_o = 6$ m/s, $p_i = 1.6 \times 10^5$ Pa, $p_a = 101325$ Pa, $T_o - T_i = 0.5$ K give $v_i = 2.4$ m/s, $F_x = 644.35$ N, $F_z = 144$ N and $\dot{Q} = 49187$ W.
- Agreement with the official and class solutions: (1), (2) and (3) are the results of CL6.pdf (which labels the vertical axis y: $F_y = \rho v_o^2 A_o$) and of the class notes.

PROBLEM CL7

Industrial Dryer: Velocities, Force, Power and Torque

Original statement · CL7.pdf

Axisymmetric | Angular momentum | Energy | Ideal gas

OFFICIAL PROBLEM STATEMENT

The axisymmetric industrial dryer of the figure in CL7.pdf is designed to provide a steady hot jet of radius R, mass flux G, and temperature T_s. Air enters the dryer from the ambient, at temperature and pressure T_a and p_a, through a radial inlet of surface A. Inside the dryer there exists an electric resistance that releases an amount of heat Q per unit time, as well as a fan of power W (work per unit time). Because of the presence of the fan, the air velocity at the exit has both azimuthal and axial components, v_θ and v_z, such that $v_\theta/v_z = \tan(\beta)$. In the analysis, assume that the fan and the inner walls of the dryer are thermally insulated, that the air behaves as an ideal gas with constant R_g and specific heat at constant pressure c_p, that gravity has a negligible effect on the fluid motion, and that the flow properties are uniform at the inlet and outlet sections. For given values of G, T_s, T_a, p_a, A, R, R_g, c_p and $\tan(\beta)$, determine:

- The velocity at the entrance section v_r as well as the two velocity components at the exit v_θ and v_z.
- The force acting on the dryer, indicating clearly its magnitude and direction.
- The total power required to run the dryer $Q + W$.
- The torque acting on the dryer.

Source: CL7.pdf (page 1). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.35, integral angular momentum balance Eq. 3.38, integral total energy balance Eq. 3.46.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady (on average) axisymmetric flow; ideal gas $p = \rho R_g T$, so $h = e + p/\rho = c_p T$; neglect $\bar{g}$; uniform properties at inlet and outlet.
- Inlet: air taken from the ambient, with $p = p_a$, $T = T_a$, $\rho_a = p_a/(R_g T_a)$ and radial velocity v_r pointing inwards (as in the official solution). Exit: free jet, $p = p_a$, $T = T_s$, $\rho_s = p_a/(R_g T_s)$, velocity $v_z \bar{e}_z + v_\theta \bar{e}_\theta$ uniform over the disc of radius R.

- G is the mass flow rate of the jet (kg/s), as used in the official solution ($G = \rho_a v_r A$). Fan and inner walls insulated; the resistance releases $\dot{Q}$; the fan delivers the power $\dot{W}$ to the air.
- Data: $[G,\ T_s,\ T_a,\ p_a,\ A,\ R,\ R_g,\ c_p,\ \tan\beta]$. Unknowns: $[v_r,\ v_z,\ v_\theta,\ \bar{F},\ \dot{Q}+\dot{W},\ \bar{M}]$.

Symbol	Meaning	SI unit
G	mass flow rate	kg/s
A, R	inlet area, jet radius	m^2, m
v_r, v_z, v_θ	inlet and exit velocities	m/s
R_g, c_p	gas constant, specific heat	J/(kg·K)
$\bar{F}$, $\bar{M}$	force and torque on the dryer	N, N·m

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the dryer; cylindrical basis $(\bar{e}_r, \bar{e}_\theta, \bar{e}_z)$ with $\bar{e}_z$ along the axis in the jet direction; origin on the axis. By axisymmetry $\int \bar{e}_r\, d\theta = \int \bar{e}_\theta\, d\theta = 0$ over a full turn.
- Surfaces: Σ_1 (radial inlet, area A, $\bar{n} = \bar{e}_r$, $\bar{v} = -v_r \bar{e}_r$), Σ_2 (exit disc, $\bar{n} = \bar{e}_z$), Σ_w (fixed casing and resistance, $\bar{v} = 0$) and Σ_m (fan blades, $\bar{v} = \bar{v}_c$).

Force and torque of the air (inside, and ambient outside) on the dryer, and the heat and work exchanged:

$$\boxed{\bar{F} = \int_{\Sigma_w, \Sigma_m} \left[(p - p_a)\, \bar{n} - \bar{\bar{\tau}}' \cdot \bar{n} \right] d\sigma}$$

$$\boxed{\bar{M} = \int_{\Sigma_w, \Sigma_m} \bar{x} \wedge \left[(p - p_a)\, \bar{n} - \bar{\bar{\tau}}' \cdot \bar{n} \right] d\sigma}$$

PHASE 3

Balances, Surface by Surface

Mass (Notes Eq. 3.11)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\,(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma$
Σ_1	$\rho_a(-v_r)A = -\rho_a v_r A$
Σ_2	$\rho_s v_z \pi R^2$
$\Sigma_w,\ \Sigma_m$	0

$$G = \frac{p_a}{R_g T_a}\,v_r A = \frac{p_a}{R_g T_s}\,v_z \pi R^2 \implies \boxed{v_r = \frac{G R_g T_a}{p_a A}, \qquad v_z = \frac{G R_g T_s}{p_a \pi R^2}, \qquad v_\theta = v_z \tan\beta} \quad (1)$$

Linear momentum (Notes Eq. 3.35)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma$	$-\int(p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_1	$\rho_a v_r^2 \int_{\Sigma_1}\bar{e}_r\,d\sigma = 0$ (axisymmetric)	0 ($p = p_a$)	0
Σ_2	$\rho_s v_z \int_{\Sigma_2}(v_z\bar{e}_z + v_\theta\bar{e}_\theta)\,d\sigma = \rho_s v_z^2 \pi R^2\,\bar{e}_z$	0 (free jet)	0
$\Sigma_w,\ \Sigma_m$	0	$-\int(p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed terms: $-\bar{F}$. On Σ_2, $\int_{\Sigma_2} \bar{e}_\theta \, d\sigma = 0$, so the swirl gives no force.

$$\rho_s v_z^2 \pi R^2 \, \bar{e}_z = -\bar{F} \implies \boxed{\bar{F} = -\frac{p_a}{R_g T_s} \, v_z^2 \, \pi R^2 \, \bar{e}_z = -G \, v_z \, \bar{e}_z} \tag{2}$$

Total energy (Notes Eq. 3.46)

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho \left(e + \frac{v^2}{2}\right) dV}_{0 \text{ (steady)}} + \int_{\Sigma_c} \rho \left(e + \frac{v^2}{2}\right) [(\bar{v} - \bar{v}_c) \cdot \bar{n}] \, d\sigma = -\int_{\Sigma_c} p \, \bar{v} \cdot \bar{n} \, d\sigma + \int_{\Sigma_c} \bar{v} \cdot (\bar{\bar{\tau}}' \cdot \bar{n}) \, d\sigma + \underbrace{\int_{V_c} \rho \bar{g} \cdot \bar{v} \, dV}_{\text{neglect } \bar{g}} - \int_{\Sigma_c} \bar{q} \cdot \bar{n} \, d\sigma + \underbrace{\int_{V_c} Q_R \, dV}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho \left(e + \frac{v^2}{2}\right) (\bar{v} \cdot \bar{n}) \, d\sigma$	$-\int p \, \bar{v} \cdot \bar{n} \, d\sigma$	$\int \bar{v} \cdot (\bar{\bar{\tau}}' \cdot \bar{n}) \, d\sigma$	$-\int \bar{q} \cdot \bar{n} \, d\sigma$
Σ_1	$-G \left(e_a + \frac{v_r^2}{2}\right)$	$+p_a v_r A$	0	0
Σ_2	$+G \left(e_s + \frac{v_\theta^2 + v_z^2}{2}\right)$	$-p_a v_z \pi R^2$	0	0
Σ_m	0	$-\int_{\Sigma_m} p \, \bar{v} \cdot \bar{n} \, d\sigma$	$\int_{\Sigma_m} \bar{v} \cdot (\bar{\bar{\tau}}' \cdot \bar{n}) \, d\sigma$	0 **(insulated)**
Σ_w	0	0	0	$-\int_{\Sigma_w} \bar{q} \cdot \bar{n} \, d\sigma$

Boxed terms: the work on Σ_m is the fan power $\dot{W}$; the heat through Σ_w (only the resistance, the walls are insulated) is $\dot{Q}$. With $p_a v_r A = G \, p_a/\rho_a$ and $p_a v_z \pi R^2 = G \, p_a/\rho_s$, $e + p/\rho = c_p T$:

$$\boxed{\dot{Q} + \dot{W} = G \left[\left(c_p T_s + \frac{v_\theta^2 + v_z^2}{2} \right) - \left(c_p T_a + \frac{v_r^2}{2} \right) \right]} \tag{3}$$

Angular momentum about the origin, z component (Notes Eq. 3.38)

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho \bar{x} \wedge \bar{v} \, dV}_{0} + \int_{\Sigma_c} \rho \, (\bar{x} \wedge \bar{v}) [(\bar{v} - \bar{v}_c) \cdot \bar{n}] \, d\sigma = -\int_{\Sigma_c} \bar{x} \wedge (p - p_a) \bar{n} \, d\sigma + \int_{\Sigma_c} \bar{x} \wedge (\bar{\bar{\tau}}' \cdot \bar{n}) \, d\sigma$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,(\bar{x}\wedge\bar{v})_z\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma$	pressure and viscous moments
Σ_1	0 ($\bar{x}\wedge(-v_r\bar{e}_r)=-zv_r\,\bar{e}_\theta$, no z component)	0
Σ_2	$\int_0^R \rho_s v_z\,(rv_\theta)\,2\pi r\,dr=\rho_s v_z v_\theta\,\frac{2\pi R^3}{3}$	0
$\Sigma_w,\ \Sigma_m$	0	$-\bar{M}\cdot\bar{e}_z$

On Σ_2, $\bar{x}=r\bar{e}_r+z_2\bar{e}_z$ and $\bar{x}\wedge(v_z\bar{e}_z+v_\theta\bar{e}_\theta)=-rv_z\bar{e}_\theta+rv_\theta\bar{e}_z-z_2v_\theta\bar{e}_r$; only $rv_\theta\bar{e}_z$ survives the integral. The antiderivative of r^2 is $r^3/3$, and Barrow's rule between 0 and R gives $R^3/3$.

$$\boxed{\bar{M}=-\rho_s v_z v_\theta\,\frac{2\pi R^3}{3}\,\bar{e}_z=-\frac{2}{3}\,R\,G\,v_\theta\,\bar{e}_z} \qquad (4)$$

Erratum in the official solution (slip in the boxed result)

CL7.pdf writes the momentum balance as $\frac{p_a}{R_gT_s}v_z^2\pi R^2\bar{e}_z=-\bar{F}$ but boxes $\bar{F}=-\frac{p_a}{R_gT_s}v_z^2\bar{e}_z$, without the area πR^2. That expression has units of pressure, not force; the correct result is (2).

PHASE 4

Interpretation, Limits and Dimensional Check

- The radial inflow gives no net momentum (axisymmetry), so the force is the reaction to the jet: magnitude Gv_z, direction $-\bar{e}_z$ (opposite to the jet).
- The swirl gives no force but gives a torque: the fan sets the air rotating, and the air reacts on the dryer with $\bar{M}$ opposite to the swirl. For $\beta=0$, $v_\theta=0$ and $\bar{M}=0$.
- (3): the electric and mechanical power go into heating the air from T_a to T_s and into its kinetic energy.
- Dimensions (SI): $[Gv_z]=\mathrm{N}$; $[RGv_\theta]=\mathrm{m}\cdot(\mathrm{kg/s})(\mathrm{m/s})=\mathrm{N}\cdot\mathrm{m}$; $[Gc_pT]=\mathrm{W}$.

- Numerical illustration (hypothetical data): $G = 0.5$ kg/s, $T_s = 350$ K, $T_a = 293$ K, $p_a = 101325$ Pa, $A = 0.05$ m^2, $R = 0.05$ m, $R_g = 287$ J/(kg K), $c_p = 1005$ J/(kg K), $\tan\beta = 0.5$ give $v_r = 8.2991$ m/s, $v_z = 63.112$ m/s, $v_\theta = 31.556$ m/s, $|\bar{F}| = 31.556$ N, $\dot{Q} + \dot{W} = 29870$ W and $|\bar{M}| = 0.52594$ N·m (a midpoint quadrature of the torque integral gives the same value).

PROBLEM CL8

Jet Impinging on a Wall with an Orifice

Original statement · CL8.pdf

Free jets | Bernoulli | Mass | Force on a wall

OFFICIAL PROBLEM STATEMENT

A circular water jet of diameter D_j and velocity V_j impinges perpendicularly to a wall, where there exists an orifice centered at the jet axis, as shown in the figure of CL8.pdf, so that part of the water flows through the orifice and part is deflected to flow next to the wall. Assuming that the flow velocity is sufficiently large for the effects of viscosity and gravity to be negligible:

- Determine the values of the velocities V_s and V_r for the secondary jet that emerges from the orifice and for the radial wall jet.
- Obtain the thickness $h(r)$ of the radial wall jet.
- Compute the force that the jet exerts on the wall as a function of ρ, V_j, D_j and D_s.

Source: CL8.pdf (page 1). Notes.pdf: Bernoulli's equation Eq. 4.17 (Chapter 4), integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, incompressible, frictionless (viscosity negligible); neglect $\bar{g}$ (buoyancy-free).
- Free jets surrounded by the ambient air at p_a. In each jet away from the wall the streamlines are straight and parallel (aligned), so the pressure does not vary across them: $p = p_a$ inside the incoming jet, the secondary jet and the wall jet (official solution).
- Data: $[\rho,\ V_j,\ D_j,\ D_s]$. Unknowns: $[V_s,\ V_r,\ h(r),\ \bar{F}_w]$.

Symbol	Meaning	SI unit
V_j, D_j	velocity and diameter of the jet	m/s, m
V_s, D_s	secondary jet velocity, orifice diameter	m/s, m
V_r, $h(r)$	wall-jet velocity and thickness	m/s, m
$\bar{F}_w$	force of the jet on the wall	N

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the wall; $\bar{e}_x$ along the jet axis (towards the wall); cylindrical coordinate r on the wall, with $\int \bar{e}_r \, d\theta = 0$.
- Fixed control volume around the water: Σ_j (section of the incoming jet, $\bar{n} = -\bar{e}_x$), Σ_s (section of the secondary jet beyond the orifice, $\bar{n} = +\bar{e}_x$), Σ_r (cylinder of radius r across the wall jet, $\bar{n} = \bar{e}_r$, height h), Σ_a (free surfaces, $p = p_a$, $\bar{v} \cdot \bar{n} = 0$) and Σ_w (wetted wall).

Force of the water (and of the ambient air on the dry side) on the wall:

$$\boxed{\bar{F}_w = \int_{\Sigma_w} (p - p_a)\, \bar{n}\, d\sigma - \int_{\Sigma_w} \bar{\bar{\tau}}' \cdot \bar{n}\, d\sigma}$$

PHASE 3

Bernoulli and Balances, Surface by Surface

3.1 Velocities: Bernoulli along a streamline

Why this tool: for steady, frictionless, incompressible flow without mass forces, $p + \frac{1}{2}\rho|\bar{v}|^2$ is constant along a streamline (Notes Eq. 4.17 with $U = 0$, as used in the official solution). Take a streamline from the incoming jet to the wall jet, and another one to the secondary jet; at both ends $p = p_a$:

$$p_a + \tfrac{1}{2}\rho V_j^2 = p_a + \tfrac{1}{2}\rho V_r^2 \implies \boxed{V_r = V_j}, \qquad p_a + \tfrac{1}{2}\rho V_j^2 = p_a + \tfrac{1}{2}\rho V_s^2 \implies \boxed{V_s = V_j} \quad (1)$$

> Mass (Notes Eq. 3.11)
>
> $$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho \, dV}}_{0 \text{ (steady)}} + \int_{\Sigma_c} \rho \left(\bar{v} - \underbrace{\cancel{\bar{v}_c}}_{0}\right) \cdot \bar{n} \, d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,\bar{v}\cdot\bar{n}\,d\sigma$
Σ_j	$-\rho V_j \frac{\pi D_j^2}{4}$
Σ_s	$+\rho V_s \frac{\pi D_s^2}{4}$
Σ_r	$+\rho V_r\, 2\pi r\, h$
$\Sigma_a,\ \Sigma_w$	0

With (1), $V_s = V_r = V_j$ and the velocity cancels.

$$-\frac{\pi D_j^2}{4} + \frac{\pi D_s^2}{4} + 2\pi r h = 0 \implies \boxed{h(r) = \frac{D_j^2 - D_s^2}{8\,r}} \qquad (2)$$

`Linear momentum (Notes Eq. 3.36)`

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho\bar{v}\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\bar{v}\,[(\bar{v} - \underbrace{\bar{v}_c}_{0})\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c}(p - p_a)\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect } \bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int (p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_j	$\rho V_j \bar{e}_x(-V_j)\frac{\pi D_j^2}{4} = -\rho V_j^2 \frac{\pi D_j^2}{4}\bar{e}_x$	$0\ (p = p_a)$	0
Σ_s	$+\rho V_s^2 \frac{\pi D_s^2}{4}\bar{e}_x$	$0\ (p = p_a)$	0
Σ_r	$\rho V_r^2 h\, r \int_0^{2\pi} \bar{e}_r\,d\theta = 0$	$0\ (p = p_a)$	0
Σ_a	0	$0\ (p = p_a)$	0
Σ_w	0	$-\int_{\Sigma_w}(p-p_a)\,\bar{n}\,d\sigma$	$\int_{\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed Σ_w terms: $-\bar{F}_w$. With (1), $V_s = V_j$.

$$\left(\rho V_j^2 \frac{\pi D_s^2}{4} - \rho V_j^2 \frac{\pi D_j^2}{4}\right) \bar{e}_x = -\bar{F}_w \implies \boxed{\bar{F}_w = \rho V_j^2 \frac{\pi}{4}\left(D_j^2 - D_s^2\right) \bar{e}_x} \quad (3)$$

PHASE 4

Interpretation, Limits and Dimensional Check

- Only the part of the jet stopped by the wall (the annulus between D_s and D_j) pushes on it; the secondary jet keeps its momentum. For $D_s \to 0$, $\bar{F}_w \to \rho V_j^2 \pi D_j^2/4$, the force of a jet on a flat plate; for $D_s \to D_j$ the jet passes through and $\bar{F}_w \to 0$, $h \to 0$.
- $h \propto 1/r$: the wall jet thins as it spreads, keeping the flow rate $2\pi r h V_j$ constant.
- Dimensions (SI): $[h] = \mathrm{m}^2/\mathrm{m} = \mathrm{m}$; $[\rho V_j^2 D^2] = \mathrm{N}$.
- Numerical illustration (hypothetical data): $\rho = 1000$ kg/m^3, $V_j = 10$ m/s, $D_j = 0.05$ m, $D_s = 0.02$ m give $|\bar{F}_w| = 164.93$ N and $h(0.1\ \mathrm{m}) = 2.625$ mm.
- Agreement with the official solution: CL8.pdf obtains (1), (2) and (3).

PROBLEM CL9

Submarine Propulsion Device: Thrust from the Propeller Power

Original statement · CL9.pdf

Thrust Bernoulli Energy Similarity

OFFICIAL PROBLEM STATEMENT

The submarine propulsion device schematically represented in the figure of CL9.pdf consists of a nozzle, with inlet and outlet cross-section area A_e and A_s, and a propeller (printed as 'fan', corrected by hand to 'propeller') located inside, which imparts a rate of work per unit time (power) $\dot{W}$ to the circulating water stream, whose density and viscosity are ρ and μ, respectively. To determine the thrust T produced when the submarine is at rest and at sufficient depth for gravity to have a negligible effect, please follow these steps:

- By application of the continuity equation determine a relationship between the uniform velocities U_e and U_s at the inlet and outlet sections.
- Use the momentum conservation equation to obtain an expression for the thrust (axial force) that the fluid exerts on the propulsion device as a function of the flow conditions at the inlet and outlet. Assume that the pressure surrounding the device is equal to the ambient pressure p_∞ .
- Assuming that the flow upstream from the entrance is steady and inviscid, use Bernoulli's equation along a given streamline to find a relationship between the flow conditions in the stagnant atmosphere (1) and at the entrance e.
- Use the energy equation to relate the work per unit time $\dot{W}$ with the different fluid variables appearing in the problem. Assume that the variation of internal energy between the inlet and outlet sections is negligible ($e_e = e_s$) and that all of the device walls, including those of the fan, are thermally insulated.
- Combine the above equations to show that the thrust can be expressed in the form $T/(\rho^{1/3}\dot{W}^{2/3}A_s^{1/3}) = f(A_e/A_s)$, obtaining the function f.

Source: CL9.pdf (page 1). Notes.pdf: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.35, integral total energy balance Eq. 3.46, Bernoulli's equation Eq. 4.17.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady (on average), incompressible water; gravity negligible at depth (it is absorbed in the reduced pressure $p + \rho g z$, as the official solution notes); uniform properties at e and s.
- Viscous stresses negligible on the sections ($\rho U D/\mu \gg 1$); the exit jet discharges at the ambient pressure, $p_s = p_\infty$; walls and propeller insulated; $e_e = e_s$.
- Data: $[\rho,\ A_e,\ A_s,\ \dot{W},\ p_\infty]$. Unknowns: $[U_e,\ U_s,\ p_e,\ T]$.

Symbol	Meaning	SI unit
A_e, A_s	inlet and outlet areas	m^2
U_e, U_s	inlet and outlet velocities	m/s
p_e, p_∞	inlet and ambient pressures	Pa
$\dot{W}$	power given by the propeller to the water	W
T	thrust	N

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the submarine (at rest), $\bar{e}_x$ along the axis in the direction of the exit jet. The thrust pushes the device towards $-\bar{e}_x$: $\bar{F} = -T\bar{e}_x$ is the force of the water (inside and outside, at p_∞) on the device.
- Surfaces: Σ_e (inlet, $\bar{n} = -\bar{e}_x$), Σ_s (outlet, $\bar{n} = +\bar{e}_x$), Σ_w (nozzle walls, fixed) and Σ_p (propeller, moving).

$$\boxed{\bar{F} = -T\,\bar{e}_x = \int_{\Sigma_w,\Sigma_p} \left[(p - p_\infty)\,\bar{n} - \bar{\bar{\tau}}' \cdot \bar{n}\right] d\sigma}$$

PHASE 3

Balances, Surface by Surface

(1) Mass (Notes Eq. 3.11)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho\,dV}}_{0\ \text{(steady)}}+\int_{\Sigma_c}\rho\,(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma=0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma$
Σ_e	$-\rho U_e A_e$
Σ_s	$+\rho U_s A_s$
$\Sigma_w,\ \Sigma_p$	0

$$\boxed{U_e A_e = U_s A_s} \qquad (1)$$

(2) Linear momentum (Notes Eq. 3.35)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}}+\int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma=-\int_{\Sigma_c}(p-p_\infty)\,\bar{n}\,d\sigma+\int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma+\underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma$	$-\int(p-p_\infty)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_e	$-\rho U_e^2 A_e\,\bar{e}_x$	$(p_e-p_\infty)A_e\,\bar{e}_x$	0
Σ_s	$+\rho U_s^2 A_s\,\bar{e}_x$	$0\ (p_s=p_\infty)$	0
$\Sigma_w,\ \Sigma_p$	0	$-\int(p-p_\infty)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed terms: $-\bar{F}=T\bar{e}_x$. With (1), $\rho U_s^2 A_s-\rho U_e^2 A_e=\rho U_e A_e(U_s-U_e)$.

$$\rho U_e A_e (U_s - U_e) = (p_e - p_\infty) A_e + T \implies \boxed{T = \rho U_e A_e \left(U_s - U_e\right) - (p_e - p_\infty) A_e} \quad (2)$$

(3) Bernoulli upstream of the inlet

Steady inviscid flow from the stagnant water far ahead (1), where $v = 0$ and $p = p_\infty$, to the inlet e (Notes Eq. 4.17 with the reduced pressure):

$$p_\infty = p_e + \tfrac{1}{2}\rho U_e^2 \implies \boxed{p_e - p_\infty = -\tfrac{1}{2}\rho U_e^2} \quad (3)$$

`(4) Total energy (Notes Eq. 3.46)`

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho\left(e + \tfrac{v^2}{2}\right) dV}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\left(e + \frac{v^2}{2}\right)\left[(\bar{v} - \bar{v}_c)\cdot\bar{n}\right] d\sigma = -\int_{\Sigma_c} p\,\bar{v}\cdot\bar{n}\, d\sigma + \int_{\Sigma_c} \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\, d\sigma + \underbrace{\int_{V_c} \rho\bar{g}\cdot\bar{v}\, dV}_{\text{neglect } \bar{g}} - \int_{\Sigma_c} \bar{q}\cdot\bar{n}\, d\sigma + \underbrace{\int_{V_c} Q_R\, dV}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e + \frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\, d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\, d\sigma$	$\int \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\, d\sigma$	$-\int \bar{q}\cdot\bar{n}\, d\sigma$
Σ_e	$-\rho U_e A_e\left(e_e + \frac{U_e^2}{2}\right)$	$+p_e U_e A_e$	0	0
Σ_s	$+\rho U_s A_s\left(e_s + \frac{U_e^2}{2}\right)$	$-p_\infty U_s A_s$	0	0
Σ_p	0	$-\int_{\Sigma_p} p\,\bar{v}\cdot\bar{n}\, d\sigma$	$\int_{\Sigma_p} \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\, d\sigma$	0 **(insulated)**
Σ_w	0	0	0	0 (insulated)

Boxed Σ_p terms: the propeller power $\dot{W}$. With $e_e = e_s$, $\rho U_e A_e = \rho U_s A_s = \dot{m}$ and (3):

$$\dot{W} = \dot{m}\left[\frac{p_\infty - p_e}{\rho} + \frac{U_s^2 - U_e^2}{2}\right] = \dot{m}\left[\frac{U_e^2}{2} + \frac{U_s^2 - U_e^2}{2}\right] \implies \boxed{\dot{W} = \tfrac{1}{2}\rho A_s U_s^3, \qquad U_s = \left(\frac{2\dot{W}}{\rho A_s}\right)^{1/3}} \quad (4)$$

(5) Thrust in similarity form

Substituting (3) into (2) and using (1), $U_e A_e = U_s A_s$ and $U_e = U_s A_s / A_e$:

$$T = \rho U_e A_e (U_s - U_e) + \tfrac{1}{2}\rho U_e^2 A_e = \rho U_e A_e \left(U_s - \frac{U_e}{2}\right) = \rho A_s U_s \left(U_s - \frac{U_s}{2}\frac{A_s}{A_e}\right) = \rho A_s U_s^2 \left(1 - \frac{1}{2}\frac{A_s}{A_e}\right).$$

With (4), $\rho A_s U_s^2 = \rho A_s \left(\frac{2\dot{W}}{\rho A_s}\right)^{2/3} = 2^{2/3}\rho^{1/3}\dot{W}^{2/3}A_s^{1/3}$, so

$$\boxed{\frac{T}{\rho^{1/3}\dot{W}^{2/3}A_s^{1/3}} = f\left(\frac{A_e}{A_s}\right) = 2^{2/3}\left(1 - \frac{1}{2\,A_e/A_s}\right)} \qquad (5)$$

PHASE 4

Interpretation, Limits and Dimensional Check

- f grows with A_e/A_s: a large intake (slow inflow, small suction depression at the inlet) gives the most thrust for a given power, tending to $f \to 2^{2/3} \approx 1.587$ as $A_e/A_s \to \infty$. For $A_e = A_s$, $f = 2^{2/3}/2 \approx 0.794$.
- Thrust grows only as $\dot{W}^{2/3}$: doubling the power gives $2^{2/3} \approx 1.59$ times the thrust, because the extra power goes into a faster jet whose kinetic energy is wasted.
- Dimensions (SI): $[\rho^{1/3}\dot{W}^{2/3}A_s^{1/3}] = (\mathrm{kg/m^3})^{1/3}(\mathrm{kg\,m^2/s^3})^{2/3}(\mathrm{m^2})^{1/3} = \mathrm{kg\cdot m/s^2} = \mathrm{N}$, so f is dimensionless.
- Numerical check (hypothetical data): $\rho = 1000$ kg/m^3, $A_s = 0.05$ m^2, $A_e = 0.1$ m^2, $\dot{W} = 50$ kW give $U_s = 12.599$ m/s, $U_e = 6.2996$ m/s, $T = 5952.8$ N, and $T/(\rho^{1/3}\dot{W}^{2/3}A_s^{1/3}) = 1.1906 = f(2)$; the energy balance (4) gives back $\dot{W} = 50$ kW.
- Agreement with the official solution: CL9.pdf obtains (1) to (5), $f = 2^{2/3}\left(1 - \tfrac{1}{2}A_s/A_e\right)$.

PROBLEM CL10

Ventilation Device: Velocities, Force and Fan Power

Original statement · CL10.pdf

Fan | Bernoulli | Linear momentum | Numerical values

OFFICIAL PROBLEM STATEMENT

The ventilation device shown in the figure of CL10.pdf takes air out of a room at pressure p_a through a circular nozzle of diameter D_e. The air is discharged to the open atmosphere, also at pressure p_a, through a pipe of diameter $D_s < D_e$. To force the air motion the device includes a fan, which exerts a rate of work per unit time $\dot{W}$ on the air stream. To simplify the solution you may assume that the air density ρ and viscosity μ are constant and that gravity has a negligible effect.

- Using the continuity equation, obtain a relationship between the values of the velocity at the entrance and exit sections U_e and U_s, assumed to be uniform.
- Make use of Bernoulli's equation upstream from the entrance section (i.e., along a stream line between (0) and (e)) to obtain an equation for the pressure and velocity at the entrance section p_e and U_e.
- Determine the two components of the force F_x and F_y acting on the device. Include in the computation the contribution of the stagnant surrounding air.
- Use the energy equation to compute the value of U_s as a function of the power $\dot{W}$. Assume that the air temperature has the same values at the entrance and exit sections and that the device is thermally insulated.
- Obtain the numerical values of U_s, U_e, p_e, F_x and F_y for $D_e = 2$ m, $D_s = 0.1$ m, $\rho = 1.2$ kg/m^3, $\mu = 1.8 \times 10^{-5}$ kg/(m·s), $\dot{W} = 1000$ W, $p_a = 101300$ Pa. Note: the following quantities are known: $D_e, D_s, \rho, \mu, \dot{W}, p_a$.

Source: CL10.pdf (page 1). Notes.pdf: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.35, integral total energy balance Eq. 3.46, Bernoulli's equation Eq. 4.17.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, constant ρ and μ; neglect $\bar{g}$; uniform properties at e and s; same temperature at e and s ($e_e = e_s$); device insulated.

- The air in the room far from the inlet is at rest at p_a (point 0); the exit is a free jet at p_a; the outer surface of the device is surrounded by stagnant air at p_a (included through the gauge pressure).
- Geometry (figure): the inlet faces downwards, air enters along $+\bar{e}_y$; the exit pipe points along $+\bar{e}_x$.
- Data: $[D_e,\ D_s,\ \rho,\ \mu,\ \dot{W},\ p_a]$. Unknowns: $[U_e,\ U_s,\ p_e,\ F_x,\ F_y]$.

Symbol	Meaning	SI unit
D_e, D_s	inlet and exit diameters	m
U_e, U_s	inlet and exit velocities	m/s
p_e, p_a	inlet and ambient pressures	Pa
$\dot{W}$	fan power given to the air	W
F_x, F_y	force on the device	N

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the device: $\bar{e}_x$ horizontal (exit direction), $\bar{e}_y$ vertical upwards (inlet flow direction).
- Surfaces: Σ_e (inlet, $\bar{n} = -\bar{e}_y$, $\bar{v} = U_e \bar{e}_y$), Σ_s (exit, $\bar{n} = +\bar{e}_x$, $\bar{v} = U_s \bar{e}_x$), Σ_w (fixed walls) and Σ_m (fan).

$$\boxed{\bar{F} = F_x \bar{e}_x + F_y \bar{e}_y = \int_{\Sigma_w, \Sigma_m} \left[(p - p_a)\, \bar{n} - \bar{\bar{\tau}}' \cdot \bar{n} \right] d\sigma}$$

PHASE 3

Balances, Surface by Surface

(1) Mass (Notes Eq. 3.11)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\,(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma$
Σ_e	$-\rho U_e \frac{\pi D_e^2}{4}$
Σ_s	$+\rho U_s \frac{\pi D_s^2}{4}$
$\Sigma_w,\ \Sigma_m$	0

$$\boxed{U_e\,D_e^2 = U_s\,D_s^2} \qquad (1)$$

(2) Bernoulli from the room (0) to the inlet (e)

$$p_a + 0 = p_e + \tfrac{1}{2}\rho U_e^2 \implies \boxed{p_e = p_a - \tfrac{1}{2}\rho U_e^2} \qquad (2)$$

(3) Linear momentum (Notes Eq. 3.35)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma$	$-\int (p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_e	$\rho U_e\bar{e}_y(-U_e)\frac{\pi D_e^2}{4} = -\rho U_e^2\frac{\pi D_e^2}{4}\bar{e}_y$	$-(p_e-p_a)(-\bar{e}_y)\frac{\pi D_e^2}{4} = (p_e-p_a)\frac{\pi D_e^2}{4}\bar{e}_y$	0
Σ_s	$+\rho U_s^2\frac{\pi D_s^2}{4}\bar{e}_x$	0 (free jet)	0
$\Sigma_w,\ \Sigma_m$	0	$-\int (p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed terms: $-\bar{F}$.

$$\boxed{F_x = -\rho U_s^2\frac{\pi D_s^2}{4}, \qquad F_y = \rho U_e^2\frac{\pi D_e^2}{4} + (p_e-p_a)\frac{\pi D_e^2}{4} = \rho U_e^2\frac{\pi D_e^2}{8}} \quad (3)$$

The last form of F_y uses (2): $(p_e - p_a) = -\frac{1}{2}\rho U_e^2$.

(4) Total energy (Notes Eq. 3.46)

$$\underbrace{\frac{d}{dt}\int_{V_c}\rho\left(e+\frac{v^2}{2}\right)dV}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\left(e+\frac{v^2}{2}\right)[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c} p\,\bar{v}\cdot\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma + \underbrace{\int_{V_c}\rho\bar{g}\cdot\bar{v}\,dV}_{\text{neglect }\bar{g}} - \int_{\Sigma_c}\bar{q}\cdot\bar{n}\,d\sigma + \underbrace{\int_{V_c}Q_R\,dV}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e+\frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	$-\int \bar{q}\cdot\bar{n}\,d\sigma$
Σ_e	$-\dot{m}\left(e_e+\frac{U_e^2}{2}\right)$	$+p_eU_e\frac{\pi D_e^2}{4}$	0	0
Σ_s	$+\dot{m}\left(e_s+\frac{U_s^2}{2}\right)$	$-p_aU_s\frac{\pi D_s^2}{4}$	0	0
Σ_m	0	$-\int_{\Sigma_m} p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int_{\Sigma_m}\bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	0
Σ_w	0	0	0	0 (insulated)

Boxed Σ_m terms: the fan power $\dot{W}$. With $e_e = e_s$, $\dot{m} = \rho U_s\pi D_s^2/4$ and (2), $\frac{p_a-p_e}{\rho} = \frac{U_e^2}{2}$:

$$\dot{W} = \dot{m}\left[\frac{p_a-p_e}{\rho} + \frac{U_s^2-U_e^2}{2}\right] = \dot{m}\,\frac{U_s^2}{2} \implies \boxed{\dot{W} = \rho\,\frac{\pi D_s^2}{8}\,U_s^3, \qquad U_s = \left(\frac{8\dot{W}}{\rho\pi D_s^2}\right)^{1/3}} \quad (4)$$

(5) Numerical values

Quantity	Expression	Value
U_s	$\left(8\dot{W}/(\rho\pi D_s^2)\right)^{1/3}$	59.647 m/s
U_e	$U_s\,(D_s/D_e)^2$	0.14912 m/s
p_e	$p_a - \frac{1}{2}\rho U_e^2$	101299.987 Pa ($p_a - p_e = 0.0133$ Pa)
F_x	$-\rho U_s^2 \pi D_s^2/4$	-33.531 N
F_y	$\rho U_e^2 \pi D_e^2/8$	0.04191 N

PHASE 4

Interpretation, Limits and Dimensional Check

- The large intake makes the inlet velocity tiny ($U_e \approx 0.15$ m/s), so the suction depression $p_a - p_e$ and F_y are negligible; the force is the reaction to the exit jet, $F_x < 0$ (opposite to the jet).
- The Reynolds numbers, $\rho U_s D_s/\mu \approx 4\times10^5$ and $\rho U_e D_e/\mu \approx 2\times10^4$, are large, which supports neglecting viscous stresses on the sections and using Bernoulli upstream of the inlet (this is where μ enters).
- Dimensions (SI): $[\rho D_s^2 U_s^3] = (\mathrm{kg/m^3})\,\mathrm{m^2(m/s)^3} = \mathrm{W}$; $[\rho U^2 D^2] = \mathrm{N}$.
- Agreement with the official solution: CL10.pdf obtains (1) to (4), with $F_x = -\rho U_s^2\pi D_s^2/4$, $F_y = \rho U_e^2\pi D_e^2/4 + (p_e - p_a)\pi D_e^2/4$ and $\dot{W} = \rho\pi D_s^2 U_s^3/8$; it does not write out the numbers of part 5.

PROBLEM CL11

VTOL Propulsion Device in Hover: Nozzle Angle and Flow State

Original statement · CL11.pdf

Thrust vectoring | Bernoulli | Linear momentum | Energy

OFFICIAL PROBLEM STATEMENT

The propulsion device of a VTOL (vertical take-off and landing) aircraft includes a horizontal intake of cross section A_i and a thrust vectoring nozzle of variable angle α and exit cross section A_e. When the aircraft is hovering (suspended at a fixed point), the engine must provide a force $\bar{F} = F_x \bar{e}_x + F_y \bar{e}_y$, such that the lift F_y equals the weight of the aircraft (Mg), while the thrust F_x is identically zero. The work per unit time on the air stream through the mobile blades is W and the amount of heat added per unit time is Q (including heat losses through the walls). For the analysis of the problem, one may introduce the following assumptions regarding the air motion:

- The effect of gravity is entirely negligible.
- Flow properties at the entrance and exit sections are uniform.
- The air around the engine casing is at rest with uniform pressure p_a.
- Upstream from the inlet the flow is inviscid and with constant density, such that $\rho_i = \rho_a$ and one may use Bernoulli's equation along any streamline to relate the ambient pressure p_a with that found at the entrance p_i.

With the above simplifications, write the equations that determine the values of U_i, U_e, p_i, ρ_e and α in terms of the known quantities ($p_a, Mg, \rho_a, A_i, A_e, \gamma, W$, and Q).

Source: CL11.pdf (page 1). Notes.pdf: integral mass balance Eq. 3.10, integral momentum balance Eq. 3.35, integral total energy balance Eq. 3.46, Bernoulli's equation Eq. 4.17.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady (on average); neglect $\bar{g}$; uniform properties at i and e; perfect gas, $h = \frac{\gamma}{\gamma-1}\frac{p}{\rho}$; ambient air at rest at p_a around the casing.
- Inlet: $\rho_i = \rho_a$ (constant density upstream) and Bernoulli from the still air to the inlet. Exit: free jet at p_a, density ρ_e.

- Geometry (figure): air enters along $-\bar{e}_x$ through the horizontal intake; the jet leaves downwards and backwards, $\bar{v}_e = -U_e(\sin\alpha\,\bar{e}_x + \cos\alpha\,\bar{e}_y)$, with α measured from the vertical.
- Hover: $F_x = 0, F_y = Mg$. Data: $[p_a,\ \rho_a,\ Mg,\ A_i,\ A_e,\ \gamma,\ \dot{W},\ \dot{Q}]$. Unknowns: $[U_i,\ U_e,\ p_i,\ \rho_e,\ \alpha]$: five equations are needed.

Symbol	Meaning	SI unit
A_i, A_e	intake and exit areas	m^2
U_i, U_e	intake and exit velocities	m/s
p_i, ρ_e	intake pressure, exit density	Pa, $\mathrm{kg/m}^3$
α	nozzle angle from the vertical	rad
$\dot{W}, \dot{Q}$	power and heat given to the air	W

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the hovering aircraft (inertial while hovering); $\bar{e}_x$ horizontal, $\bar{e}_y$ vertical upwards.
- Surfaces: Σ_i (intake, $\bar{n} = +\bar{e}_x, \bar{v} = -U_i\bar{e}_x$), Σ_e (exit, $\bar{n} = \bar{v}_e/U_e$), Σ_w (fixed walls) and Σ_m (mobile blades).

$$\boxed{\bar{F} = F_x\bar{e}_x + F_y\bar{e}_y = \int_{\Sigma_w,\Sigma_m} \left[(p - p_a)\,\bar{n} - \bar{\bar{\tau}}' \cdot \bar{n}\right] d\sigma} \qquad \text{(force of the air on the engine)}$$

PHASE 3

Balances, Surface by Surface

Mass (Notes Eq. 3.10)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho\, dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\,(\bar{v} - \bar{v}_c)\cdot\bar{n}\, d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma$
Σ_i	$\rho_a(-U_i)A_i$
Σ_e	$+\rho_e U_e A_e$
Σ_w, Σ_m	0

$$\boxed{\rho_a U_i A_i = \rho_e U_e A_e = \dot{m}} \tag{1}$$

Entrance region (Notes Eq. 4.17, from the still ambient air to the inlet):

$$\boxed{p_a = p_i + \tfrac{1}{2}\rho_a U_i^2} \tag{2}$$

`Linear momentum (Notes Eq. 3.35)`

$$\underbrace{\frac{d}{dt}\int_{V_c}\rho\bar{v}\,dV}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\int_{V_c}\rho\bar{g}\,dV}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma$	$-\int(p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_i	$\rho_a(-U_i\bar{e}_x)(-U_i)A_i = \rho_a U_i^2 A_i\,\bar{e}_x$	$-(p_i-p_a)A_i\,\bar{e}_x$	0
Σ_e	$-\rho_e U_e^2 A_e(\sin\alpha\,\bar{e}_x + \cos\alpha\,\bar{e}_y)$	0 (free jet)	0
Σ_w, Σ_m	0	$-\int(p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed terms: $-\bar{F} = -F_x\bar{e}_x - F_y\bar{e}_y$. x component with $F_x = 0$, (2) and (1): $\rho_a U_i^2 A_i - \rho_e U_e^2 A_e \sin\alpha = \tfrac{1}{2}\rho_a U_i^2 A_i$, i.e. $\tfrac{1}{2}\rho_a U_i^2 A_i = \rho_a U_i A_i\, U_e \sin\alpha$. y component with $F_y = Mg$: $-\rho_e U_e^2 A_e \cos\alpha = -Mg$.

$$\boxed{U_i = 2\,U_e \sin\alpha} \quad (3) \qquad \boxed{Mg = \rho_e U_e^2 A_e \cos\alpha = \rho_a U_i^2 A_i \,\frac{\cos\alpha}{2\sin\alpha}} \quad (4)$$

`Total energy (Notes Eq. 3.46)`

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho\left(e + \tfrac{v^2}{2}\right) dV}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\left(e + \frac{v^2}{2}\right)\left[(\bar{v} - \bar{v}_c)\cdot\bar{n}\right] d\sigma = -\int_{\Sigma_c} p\,\bar{v}\cdot\bar{n}\,d\sigma + \int_{\Sigma_c} \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma + \underbrace{\int_{V_c} \rho\bar{g}\cdot\bar{v}\,dV}_{\text{neglect } \bar{g}} - \int_{\Sigma_c} \bar{q}\cdot\bar{n}\,d\sigma + \int_{V_c} Q_R\,dV$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e + \frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	$-\int \bar{q}\cdot\bar{n}\,d\sigma$
Σ_i	$-\dot{m}\left(e_i + \frac{U_i^2}{2}\right)$	$+p_i U_i A_i$	0	0
Σ_e	$+\dot{m}\left(e_e + \frac{U_e^2}{2}\right)$	$-p_a U_e A_e$	0	0
Σ_m	0	$-\int_{\Sigma_m} p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int_{\Sigma_m} \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	$-\int_{\Sigma_m} \bar{q}\cdot\bar{n}\,d\sigma$
Σ_w	0	0	0	$-\int_{\Sigma_w} \bar{q}\cdot\bar{n}\,d\sigma$

Boxed terms: the work on Σ_m is $\dot{W}$; the heat through the walls plus the volume term $\int_{V_c} Q_R\,dV$ is $\dot{Q}$ (net, including the losses). With $h = e + p/\rho = \frac{\gamma}{\gamma-1}\frac{p}{\rho}$:

$$\boxed{\rho_a U_i A_i \left[\left(\frac{\gamma}{\gamma-1}\frac{p_a}{\rho_e} + \frac{U_e^2}{2}\right) - \left(\frac{\gamma}{\gamma-1}\frac{p_i}{\rho_a} + \frac{U_i^2}{2}\right)\right] = \dot{W} + \dot{Q}} \quad (5)$$

(1) to (5) are five equations for U_i, U_e, p_i, ρ_e and α, which answers the question.

PHASE 4

Interpretation, Limits and Dimensional Check

- (3) says that the backward component of the jet momentum must cancel the intake momentum plus the suction force at the intake: because the intake is at a lower pressure than ambient (2), only half of $\rho_a U_i^2 A_i$ has to be balanced.

- (4): the lift is the vertical momentum flux of the jet. For $\alpha \to 0$ (vertical jet), (3) would require $U_i \to 0$: the intake momentum could not be balanced, so a hovering engine with a horizontal intake needs $\alpha > 0$.
- Dimensions (SI): $[\rho_e U_e^2 A_e] = \mathrm{N} = [Mg]$; $[\dot{m}\,\gamma p/((\gamma-1)\rho)] = \mathrm{W}$.
- Numerical illustration (hypothetical data, solved by Newton's method on the two equations left after eliminating U_i, p_i, ρ_e): $\rho_a = 1.2$ kg/m^3, $p_a = 101325$ Pa, $A_i = 2.5$ m^2, $A_e = 1.2$ m^2, $Mg = 80$ kN, $\dot{W} + \dot{Q} = 6$ MW, $\gamma = 1.4$ give $U_e = 212.22$ m/s, $\alpha = 71.85^\circ$, $U_i = 403.32$ m/s, $p_i = 3727.19$ Pa, $\rho_e = 4.7511$ kg/m^3 (residuals below 10^{-6}).
- Agreement with the official solution: CL11.pdf obtains (1) to (5).

PROBLEM CL12

Gas Discharging through an Inverted U-Pipe: Outlet State, Force and Moment

Original statement · CL12.pdf

Compressible | Energy | Linear momentum | Angular momentum

OFFICIAL PROBLEM STATEMENT

The gas circulating along the pipe of cross section A depicted in the figure of CL12.pdf discharges to the atmosphere, where the pressure is p_a. Measurements at a given upstream location give V_u, p_u, and T_u for the velocity, pressure and temperature, all uniform across the section. In the analysis that follows, it is assumed that the pipe is thermally insulated and that buoyancy forces have a negligible effect on the gas motion. The gas properties are defined by its gas constant R_g together with the ratio of specific heats $\gamma = c_p/c_v$.

- Obtain the values of the gas temperature and velocity at the outlet section T_o and V_o in terms of V_u, p_u, T_u, p_a, R_g and γ.
- Compute the two components of the force F_x and F_y acting on the stretch of pipe considered.
- Calculate the momentum on the pipe with respect to the origin of the coordinate system indicated in the figure.

Source: CL12.pdf (page 1, exam of Dec 16, 2011). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral angular momentum balance Eq. 3.38, integral total energy balance Eq. 3.46.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady compressible flow of a perfect gas ($p = \rho R_g T$, $h = c_p T = \frac{\gamma}{\gamma-1} R_g T$); neglect $\bar{g}$; pipe insulated and at rest (no heat, no work); uniform properties at both sections.
- Geometry (figure): inverted U of constant section A; the gas goes up the left leg ($\bar{v} = V_u \bar{e}_y$ at the measuring section) and down the right leg, discharging as a free jet ($p = p_a$, $\bar{v} = -V_o \bar{e}_y$) at a horizontal distance L from the left leg; the origin O is on the axis of the left leg.
- Data: $[V_u,\ p_u,\ T_u,\ p_a,\ R_g,\ \gamma,\ A,\ L]$. Unknowns: $[T_o,\ V_o,\ F_x,\ F_y,\ \bar{M}]$. The 'momentum on the pipe' of the statement is the moment of the forces on the pipe about O.

Symbol	Meaning	SI unit
V_u, p_u, T_u	upstream state	m/s, Pa, K
V_o, T_o	outlet velocity and temperature	m/s, K
A, L	pipe section, distance between legs	m^2, m
$\bar{F}, \bar{M}$	force and moment on the pipe	N, N·m

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the pipe; $\bar{e}_x$ horizontal (from the left leg to the right leg), $\bar{e}_y$ vertical upwards, $\bar{e}_z = \bar{e}_x \wedge \bar{e}_y$ towards the reader. Fixed control volume.
- Surfaces: Σ_u (measuring section, $\bar{n} = -\bar{e}_y$), Σ_o (outlet, $\bar{n} = -\bar{e}_y$), Σ_w (inner wall). Positions: $\bar{x} = y\bar{e}_y + \bar{\eta}$ on Σ_u and $\bar{x} = L\bar{e}_x + y\bar{e}_y + \bar{\eta}$ on Σ_o, with $\int \bar{\eta}\, d\sigma = 0$.

$$\boxed{\bar{F} = \int_{\Sigma_w} \left[(p - p_a)\, \bar{n} - \bar{\bar{\tau}}' \cdot \bar{n} \right] d\sigma}$$

$$\boxed{\bar{M} = \int_{\Sigma_w} \bar{x} \wedge \left[(p - p_a)\, \bar{n} - \bar{\bar{\tau}}' \cdot \bar{n} \right] d\sigma}$$

PHASE 3

Balances, Surface by Sheet

Mass (Notes Eq. 3.11)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho\, dV}}_{0 \text{ (steady)}} + \int_{\Sigma_c} \rho \left(\bar{v} - \underbrace{\cancel{\bar{v}_c}}_{0} \right) \cdot \bar{n}\, d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\, \bar{v} \cdot \bar{n}\, d\sigma$
Σ_u	$-\rho_u V_u A$, with $\rho_u = \frac{p_u}{R_g T_u}$
Σ_o	$+\rho_o V_o A$, with $\rho_o = \frac{p_a}{R_g T_o}$
Σ_w	0

$$\rho_u V_u A = \rho_o V_o A \implies \boxed{\frac{p_u V_u}{T_u} = \frac{p_a V_o}{T_o}} \qquad (1)$$

`Total energy (Notes Eq. 3.46)`

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho\left(e + \tfrac{v^2}{2}\right) dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\left(e + \frac{v^2}{2}\right)\left[(\bar{v} - \underbrace{\bar{v}_c}_{0}) \cdot \bar{n}\right] d\sigma = -\int_{\Sigma_c} p\, \bar{v} \cdot \bar{n}\, d\sigma + \int_{\Sigma_c} \bar{v} \cdot (\bar{\bar{\tau}}' \cdot \bar{n})\, d\sigma + \underbrace{\cancel{\int_{V_c} \rho \bar{g} \cdot \bar{v}\, dV}}_{\text{neglect } \bar{g}} - \int_{\Sigma_c} \bar{q} \cdot \bar{n}\, d\sigma + \underbrace{\cancel{\int_{V_c} Q_R\, dV}}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e + \frac{v^2}{2}\right)(\bar{v} \cdot \bar{n})\, d\sigma$	$-\int p\, \bar{v} \cdot \bar{n}\, d\sigma$	$\int \bar{v} \cdot (\bar{\bar{\tau}}' \cdot \bar{n})\, d\sigma$	$-\int \bar{q} \cdot \bar{n}\, d\sigma$
Σ_u	$-\dot{m}\left(e_u + \frac{V_u^2}{2}\right)$	$+p_u V_u A$	0	0
Σ_o	$+\dot{m}\left(e_o + \frac{V_o^2}{2}\right)$	$-p_a V_o A$	0	0
Σ_w	0	0 ($\bar{v} = 0$)	0 ($\bar{v} = 0$)	0 (insulated)

$$\boxed{\frac{\gamma}{\gamma - 1} R_g T_u + \frac{1}{2} V_u^2 = \frac{\gamma}{\gamma - 1} R_g T_o + \frac{1}{2} V_o^2} \qquad (2)$$

Solving (1) and (2): with $C \equiv \frac{p_u V_u}{p_a T_u}$, (1) gives $V_o = C\, T_o$; substituting into (2), $\frac{C^2}{2} T_o^2 + \frac{\gamma R_g}{\gamma - 1} T_o - H = 0$ with $H = \frac{\gamma R_g}{\gamma - 1} T_u + \frac{V_u^2}{2}$, whose positive root is

$$\boxed{T_o = \frac{-\frac{\gamma R_g}{\gamma-1} + \sqrt{\left(\frac{\gamma R_g}{\gamma-1}\right)^2 + 2C^2 H}}{C^2}, \qquad V_o = C\,T_o} \tag{3}$$

Linear momentum (Notes Eq. 3.36)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho\bar{v}\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\bar{v}\,[(\bar{v} - \underbrace{\bar{v}_c}_{0})\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int(p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_u	$\rho_u V_u \bar{e}_y(-V_u)A = -\rho_u V_u^2 A\,\bar{e}_y$	$-(p_u-p_a)(-\bar{e}_y)A = (p_u-p_a)A\,\bar{e}_y$	0
Σ_o	$\rho_o(-V_o\bar{e}_y)(V_o)A = -\rho_u V_u A\,V_o\,\bar{e}_y$	0 (free jet)	0
Σ_w	0	$-\int_{\Sigma_w}(p-p_a)\,\bar{n}\,d\sigma$	$\int_{\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed Σ_w terms: $-\bar{F}$. The Σ_o entry uses (1).

$$-\rho_u V_u A\,(V_u+V_o)\,\bar{e}_y = (p_u-p_a)A\,\bar{e}_y - \bar{F} \implies \boxed{F_x = 0, \qquad F_y = \rho_u V_u A\,(V_u+V_o) + (p_u-p_a)A} \tag{4}$$

Angular momentum about O (Notes Eq. 3.38)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho\bar{x}\wedge\bar{v}\,dV}}_{0} + \int_{\Sigma_c}\rho\,(\bar{x}\wedge\bar{v})(\bar{v}\cdot\bar{n})\,d\sigma = -\int_{\Sigma_c}\bar{x}\wedge(p-p_a)\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,(\bar{x}\wedge\bar{v})(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int\bar{x}\wedge(p-p_a)\bar{n}\,d\sigma$	$\int\bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$
Σ_u	0 ($y\bar{e}_y\wedge\bar{e}_y = 0$, $\int\bar{\eta}\,d\sigma = 0$)	0 (same reason)	0

Surface	$\int \rho\,(\bar{x}\wedge\bar{v})(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int \bar{x}\wedge(p-p_a)\bar{n}\,d\sigma$	$\int \bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$
Σ_o	$\rho_o V_o A\,(L\bar{e}_x\wedge(-V_o\bar{e}_y)) = -\rho_u V_u A\,LV_o\,\bar{e}_z$	0 (free jet)	0
Σ_w	0	$-\int_{\Sigma_w}\bar{x}\wedge(p-p_a)\bar{n}\,d\sigma$	$\int_{\Sigma_w}\bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$

Boxed Σ_w terms: $-\bar{M}$.

$$-\rho_u V_u A\,LV_o\,\bar{e}_z = -\bar{M} \implies \boxed{\bar{M} = \rho_u V_u A\,L\,V_o\,\bar{e}_z = \frac{p_u}{R_g T_u} V_u A\,L\,V_o\,\bar{e}_z} \quad (5)$$

PHASE 4

Interpretation, Limits and Dimensional Check

- Adiabatic, workless flow keeps the stagnation enthalpy (2); the pressure drop to p_a lowers the density, so the gas accelerates ($V_o > V_u$) and cools slightly ($T_o < T_u$).
- Both legs push the pipe upwards: the gas turning down in the right leg and the gauge pressure at the measuring section. The moment comes only from the outlet jet, offset by L from O.
- Incompressible-like limit ($p_u \to p_a$, small V): $T_o \to T_u$, $V_o \to V_u$, $F_y \to 2\rho_u V_u^2 A$.
- Dimensions (SI): $[\rho_u V_u^2 A] = \mathrm{N}$; $[\rho_u V_u A L V_o] = \mathrm{N\cdot m}$; $[R_g T] = [V^2] = \mathrm{m^2/s^2}$.
- Numerical illustration (hypothetical data, air): $V_u = 50$ m/s, $p_u = 1.5\times10^5$ Pa, $T_u = 400$ K, $p_a = 101325$ Pa, $R_g = 287$ J/(kg K), $\gamma = 1.4$, $A = 0.01$ m^2, $L = 1$ m give $T_o = 398.537$ K, $V_o = 73.749$ m/s, $F_y = 567.6$ N, $M_z = 48.181$ N·m (residuals of (1), (2) below 10^{-8}).
- Agreement with the official solution: CL12.pdf obtains (1), (2), (4) and (5); it does not solve (1)-(2) explicitly.

PROBLEM CL13

Centrifugal Pump: Exit Velocity, Axial Force, Torque and Power

Original statement · CL13.pdf

Turbomachine | Angular momentum | Euler torque | Pump power

OFFICIAL PROBLEM STATEMENT

The figure of CL13.pdf shows a sketch of a centrifugal pump. Water enters the pump with uniform axial velocity V_o and pressure p_o through a circular pipe of diameter D_o. The internal geometry of the pump includes rotating blades that redirect the flow, so that the water leaves the pump with zero axial velocity through a lateral exit of width h and diameter D. At the exit section, the pressure is $p_s \neq p_a$ and the velocity vector, of magnitude V_s, forms an angle α with the lateral surface, as indicated in the figure, i.e., the corresponding normal and tangential velocity components are $V_s \sin\alpha$ and $V_s \cos\alpha$, respectively. Assuming that the effect of gravity is negligible and that the water temperature at the exit is equal to that found at the entrance (handwritten addition: and that the pump walls are insulated), determine the values of the:

- Radial velocity component at the exit section $V_s \sin\alpha$.
- Axial force exerted on the pump.
- Torque exerted on the pump.
- Power of the pump, assumed to be ideal (i.e., work per unit time exerted on the water stream by the rotating parts).

Note: besides the water density ρ and the ambient pressure p_a, the following quantities can be assumed to be known in solving the problem: D, D_o, h, α, V_o, p_o, and p_s.

Source: CL13.pdf (page 1, exam of Jan 10, 2012). Notes.pdf, Chapter 3: integral mass balance Eq. 3.10, integral momentum balance Eq. 3.35, integral angular momentum balance Eq. 3.38, integral total energy balance Eq. 3.46.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady (on average), incompressible water; neglect $\bar{g}$; equal temperature at entrance and exit ($e_s = e_o$); insulated walls; uniform properties at both sections.
- The outer casing is surrounded by air at p_a (gauge pressure $p - p_a$ in the force).
- Data: $[\rho,\ p_a,\ D,\ D_o,\ h,\ \alpha,\ V_o,\ p_o,\ p_s]$. Unknowns: $[V_s \sin\alpha,\ F_x,\ \bar{M},\ \dot{W}]$.

Symbol	Meaning	SI unit
D_o, V_o, p_o	inlet diameter, velocity, pressure	m, m/s, Pa
D, h	exit diameter and width	m
V_s, α, p_s	exit velocity, angle, pressure	m/s, rad, Pa
$\dot{W}$	power given to the water by the blades	W

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the casing; $\bar{e}_x$ along the axis (inlet flow direction); cylindrical basis $(\bar{e}_r, \bar{e}_\theta)$ around $\bar{e}_x$ with $\bar{e}_r \wedge \bar{e}_\theta = \bar{e}_x$; origin on the axis in the exit plane. Over a full turn, $\int \bar{e}_r \, d\theta = \int \bar{e}_\theta \, d\theta = 0$.
- Surfaces: Σ_o (inlet disc, $\bar{n} = -\bar{e}_x$), Σ_s (lateral exit cylinder, radius $D/2$, width h, $\bar{n} = \bar{e}_r$, $\bar{v} = V_s \sin\alpha \, \bar{e}_r + V_s \cos\alpha \, \bar{e}_\theta$), Σ_w (fixed casing) and Σ_b (rotating blades).

$$\boxed{\bar{F} = \int_{\Sigma_w, \Sigma_b} \left[(p - p_a)\, \bar{n} - \bar{\bar{\tau}}' \cdot \bar{n} \right] d\sigma}$$

$$\boxed{\bar{M} = \int_{\Sigma_w, \Sigma_b} \bar{x} \wedge \left[(p - p_a)\, \bar{n} - \bar{\bar{\tau}}' \cdot \bar{n} \right] d\sigma}$$

PHASE 3

Balances, Surface by Sheet

Mass (Notes Eq. 3.10)

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho \, dV}_{0 \text{ (steady)}} + \int_{\Sigma_c} \rho \left(\bar{v} - \bar{v}_c \right) \cdot \bar{n} \, d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,(\bar{v}-\bar{v}_c)\cdot\bar{n}\,d\sigma$
Σ_o	$-\rho V_o \frac{\pi D_o^2}{4}$
Σ_s	$+\rho V_s \sin\alpha\, \pi D h$
$\Sigma_w,\ \Sigma_b$	0

$$\boxed{V_s \sin\alpha = \frac{D_o^2}{4Dh}\,V_o}, \qquad \dot{m} = \rho V_o \frac{\pi D_o^2}{4} \tag{1}$$

`Linear momentum, axial component (Notes Eq. 3.35)`

$$\underbrace{\frac{d}{dt}\int_{V_c}\rho\bar{v}\,dV}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\int_{V_c}\rho\bar{g}\,dV}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho v_x\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma$	$-\int (p-p_a)\,n_x\,d\sigma$	$\int (\bar{\bar{\tau}}'\cdot\bar{n})_x\,d\sigma$
Σ_o	$-\rho V_o^2 \frac{\pi D_o^2}{4}$	$+(p_o-p_a)\frac{\pi D_o^2}{4}$	0
Σ_s	$0\ (v_x=0)$	$0\ (n_x=0)$	0
$\Sigma_w,\ \Sigma_b$	0	$-\int (p-p_a)\,n_x\,d\sigma$	$\int (\bar{\bar{\tau}}'\cdot\bar{n})_x\,d\sigma$

Boxed terms: $-F_x$. The radial and azimuthal components vanish because $\int \bar{e}_r\,d\theta = \int \bar{e}_\theta\,d\theta = 0$.

$$\boxed{\bar{F} = \frac{\pi D_o^2}{4}\left[\rho V_o^2 + (p_o - p_a)\right]\bar{e}_x} \tag{2}$$

`Angular momentum about the axis (Notes Eq. 3.38)`

$$\underbrace{\frac{d}{dt}\int_{V_c}\rho\,\bar{x}\wedge\bar{v}\,dV}_{0}+\int_{\Sigma_c}\rho\,(\bar{x}\wedge\bar{v})[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma=-\int_{\Sigma_c}\bar{x}\wedge(p-p_a)\bar{n}\,d\sigma+\int_{\Sigma_c}\bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int\rho\,(\bar{x}\wedge\bar{v})_x\,[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma$	pressure and viscous moments
Σ_o	0 ($\bar{v}\parallel\bar{e}_x$, no swirl)	0
Σ_s	$\rho V_s\sin\alpha\,\pi Dh\,\frac{D}{2}V_s\cos\alpha=\dot{m}\,\frac{D}{2}\,V_s\cos\alpha$	0 ($\bar{x}\wedge\bar{e}_r$ has no x component)
$\Sigma_w,\ \Sigma_b$	0	$-\bar{M}\cdot\bar{e}_x$

On Σ_s, $\bar{x}=\frac{D}{2}\bar{e}_r+x\bar{e}_x$, and the x component of $\bar{x}\wedge\bar{v}$ is $\frac{D}{2}V_s\cos\alpha$ (from $\bar{e}_r\wedge\bar{e}_\theta=\bar{e}_x$). The mass flux in front of it is (1).

$$\boxed{\bar{M}=-\rho V_o\frac{\pi D_o^2}{4}\,V_s\cos\alpha\,\frac{D}{2}\,\bar{e}_x}\qquad(3)$$

Total energy (Notes Eq. 3.46)

$$\underbrace{\frac{d}{dt}\int_{V_c}\rho\left(e+\tfrac{v^2}{2}\right)dV}_{0\text{ (steady)}}+\int_{\Sigma_c}\rho\left(e+\frac{v^2}{2}\right)[(\bar{v}-\bar{v}_c)\cdot\bar{n}]\,d\sigma=-\int_{\Sigma_c}p\,\bar{v}\cdot\bar{n}\,d\sigma+\int_{\Sigma_c}\bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma+\underbrace{\int_{V_c}\rho\bar{g}\cdot\bar{v}\,dV}_{\text{neglect }\bar{g}}-\int_{\Sigma_c}\bar{q}\cdot\bar{n}\,d\sigma+\underbrace{\int_{V_c}Q_R\,dV}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int\rho\left(e+\frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int\bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	$-\int\bar{q}\cdot\bar{n}\,d\sigma$
Σ_o	$-\dot{m}\left(e_o+\frac{V_o^2}{2}\right)$	$+p_oV_o\frac{\pi D_o^2}{4}$	0	0
Σ_s	$+\dot{m}\left(e_s+\frac{V_s^2}{2}\right)$	$-p_sV_s\sin\alpha\,\pi Dh$	0	0
Σ_b	0	$-\int_{\Sigma_b}p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int_{\Sigma_b}\bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	0
Σ_w	0	0	0	0 (insulated)

Boxed Σ_b terms: the power $\dot{W}$ of the blades. With $e_s=e_o$ and (1):

$$\boxed{\dot{W} = \rho V_o \frac{\pi D_o^2}{4}\left[\left(\frac{p_s}{\rho} + \frac{V_s^2}{2}\right) - \left(\frac{p_o}{\rho} + \frac{V_o^2}{2}\right)\right]} \qquad (4)$$

PHASE 4

Interpretation, Limits and Dimensional Check

- (3) is the reaction to the angular momentum given to the water: the blades give the water the swirl $V_s \cos\alpha$ at radius $D/2$, and the water acts on the pump with the opposite torque (Euler's turbomachinery relation; the motor must supply it).
- (4): the ideal power raises the total pressure $p + \frac{1}{2}\rho V^2$ of the water flow rate $\dot{m}/\rho$.
- Dimensions (SI): $[\dot{m}\, D\, V_s] = \mathrm{N}\cdot\mathrm{m}$; $[\dot{m}\, p/\rho] = \mathrm{W}$; $[D_o^2 V_o/(Dh)] = \mathrm{m/s}$.
- Numerical illustration (hypothetical data): $\rho = 1000\ \mathrm{kg/m^3}$, $D = 0.3\ \mathrm{m}$, $D_o = 0.1\ \mathrm{m}$, $h = 0.02\ \mathrm{m}$, $\alpha = 30^\circ$, $V_o = 3\ \mathrm{m/s}$, $p_o = 1.2 \times 10^5\ \mathrm{Pa}$, $p_s = 2.5 \times 10^5\ \mathrm{Pa}$, $p_a = 101325\ \mathrm{Pa}$ give $V_s \sin\alpha = 1.25\ \mathrm{m/s}$ ($V_s = 2.5\ \mathrm{m/s}$), $F_x = 217.36\ \mathrm{N}$, $|\bar{M}| = 7.652\ \mathrm{N{\cdot}m}$, $\dot{W} = 3030.7\ \mathrm{W}$.
- Agreement with the official solution: CL13.pdf obtains (1) to (4).

PROBLEM CL14

Gas-Turbine Diffuser Module: Exit State, Force and Torque

Original statement · CL14.pdf

Stator | Swirl removal | Angular momentum | Perfect gas

OFFICIAL PROBLEM STATEMENT

The diffuser module at the exit of a gas turbine consists of a **fixed** aerofoil cascade (stator), of inner and outer radii R_i and R_o, respectively, together with a conical diffuser, as depicted in the figure of CL14.pdf. The gas enters the module with velocity field $\bar{v}_1 = u_1\bar{e}_z + u_\theta\bar{e}_\theta$, having both axial and azimuthal components, pressure p_1 and density ρ_1, and leaves the module with a purely axial velocity field $\bar{v}_2 = u_2\bar{e}_z$, pressure p_2 and density ρ_2. For the analysis of the problem, one may introduce the following assumptions:

- The effect of gravity is entirely negligible.
- Flow properties at the entrance and exit sections are uniform.
- The air around the casing is at rest with uniform pressure p_a.
- The working gas behaves as a perfect gas (gas constant R_g, specific heat ratio γ).
- All the inner walls are adiabatic.

With the above simplifications, write the equations that determine the values of ρ_2 and u_2, as well as the net force, F, and torque, M, acting on the diffuser module, in terms of the known quantities (p_a, p_1, ρ_1, p_2, u_1, u_θ, R_i, R_o, γ and R_g).

Source: CL14.pdf (pages 1-4, quiz of Nov 5, 2012, with typed solution). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral angular momentum balance Eq. 3.38, integral total energy balance Eq. 3.46.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, axisymmetric; neglect $\bar{g}$; uniform properties at both sections; perfect gas, $h = c_pT = \frac{\gamma}{\gamma-1}\frac{p}{\rho}$; adiabatic walls; the module is fixed (no work).
- Inlet: annulus $R_i < r < R_o$ (area $A_1 = \pi(R_o^2 - R_i^2)$). Exit: full disc $r < R_o$ (area $A_2 = \pi R_o^2$).

- The ambient air at p_a touches only the outer cylindrical casing, whose normal is radial: by axisymmetry it gives no axial force and no axial torque ($\bar{F}_{ext} = 0$, $\bar{M}_{ext} = 0$). Absolute pressures are therefore used, as in the official solution (the gauge form would wrongly add $p_a \pi R_i^2$ on the hub face, which is not in contact with the ambient air).
- Data: $[p_a,\ p_1,\ \rho_1,\ p_2,\ u_1,\ u_\theta,\ R_i,\ R_o,\ \gamma,\ R_g]$. Unknowns: $[\rho_2,\ u_2,\ F,\ M]$.

Symbol	Meaning	SI unit
u_1, u_θ	axial and azimuthal inlet velocity	m/s
p_1, ρ_1; p_2, ρ_2	inlet and exit states	Pa, kg/m^3
u_2	exit axial velocity	m/s
R_i, R_o	inner and outer radii	m
F, M	axial force and torque on the module	N, N·m

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the module; right-handed cylindrical basis $(\bar{e}_r, \bar{e}_\theta, \bar{e}_z)$, $\bar{e}_r \wedge \bar{e}_\theta = \bar{e}_z$, with $\bar{e}_z$ along the flow; origin on the axis.
- Surfaces (as in the official solution): Σ_{in} (inlet annulus, $\bar{n} = -\bar{e}_z$), Σ_{out} (exit disc, $\bar{n} = +\bar{e}_z$), Σ_w (inner wall of the casing and outer wall of the cone) and Σ_a (surfaces around the stator aerofoils). All are fixed.

$$\boxed{F\,\bar{e}_z = \int_{\Sigma_w,\Sigma_a} \left[p\,\bar{n} - \bar{\bar{\tau}}'\cdot\bar{n}\right] d\sigma + \underbrace{\bar{F}_{ext}}_{0}}$$

$$\boxed{M\,\bar{e}_z = \int_{\Sigma_w,\Sigma_a} \bar{x}\wedge\left[p\,\bar{n} - \bar{\bar{\tau}}'\cdot\bar{n}\right] d\sigma + \underbrace{\bar{M}_{ext}}_{0}}$$

PHASE 3

Balances, Surface by Surface

Mass (Notes Eq. 3.11)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\,dV}}_{0\ \text{(steady)}}+\int_{\Sigma_c}\rho\,(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}\,d\sigma=0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,\bar{v}\cdot\bar{n}\,d\sigma$
Σ_{in}	$-\rho_1 u_1 \pi (R_o^2 - R_i^2)$
Σ_{out}	$+\rho_2 u_2 \pi R_o^2$
$\Sigma_w,\ \Sigma_a$	0

$$\boxed{\rho_1 u_1 \pi (R_o^2 - R_i^2) = \rho_2 u_2 \pi R_o^2 = G} \tag{1}$$

Linear momentum, axial component (Notes Eq. 3.36)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}}+\int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}]\,d\sigma=-\int_{\Sigma_c}p\,\bar{n}\,d\sigma+\int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma+\underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho v_z\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,n_z\,d\sigma$	$\int(\bar{\bar{\tau}}'\cdot\bar{n})_z\,d\sigma$
Σ_{in}	$-\rho_1 u_1^2 \pi (R_o^2 - R_i^2)$	$+p_1 \pi (R_o^2 - R_i^2)$	0 (uniform)
Σ_{out}	$+\rho_2 u_2^2 \pi R_o^2$	$-p_2 \pi R_o^2$	0 (uniform)
$\Sigma_w,\ \Sigma_a$	0	$-\int p\,n_z\,d\sigma$	$\int(\bar{\bar{\tau}}'\cdot\bar{n})_z\,d\sigma$

Boxed terms: the force of the walls on the gas, $-F$ (with $\bar{F}_{ext} = 0$). The swirl gives no axial momentum flux.

$$\boxed{F = \pi(R_o^2 - R_i^2)\left(p_1 + \rho_1 u_1^2\right) - \pi R_o^2\left(p_2 + \rho_2 u_2^2\right)} \tag{2}$$

Angular momentum, axial component (Notes Eq. 3.38)

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho\,\bar{x}\wedge\bar{v}\,dV}_{0} + \int_{\Sigma_c} \rho\,(\bar{x}\wedge\bar{v})(\bar{v}\cdot\bar{n})\,d\sigma = -\int_{\Sigma_c} \bar{x}\wedge p\,\bar{n}\,d\sigma + \int_{\Sigma_c} \bar{x}\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,(\bar{x}\wedge\bar{v})_z\,(\bar{v}\cdot\bar{n})\,d\sigma$	pressure and viscous moments
Σ_{in}	$\int_{R_i}^{R_o} \rho_1\,(r u_\theta)(-u_1)\,2\pi r\,dr = -\frac{2\pi}{3}\rho_1 u_1 u_\theta (R_o^3 - R_i^3)$	0 ($\bar{x}\wedge\bar{e}_z$ has no z component)
Σ_{out}	0 (no swirl)	0
$\Sigma_w,\ \Sigma_a$	0	$-M$

On Σ_{in}, $\bar{x} = r\bar{e}_r + z\bar{e}_z$ and the z component of $\bar{x}\wedge(u_1\bar{e}_z + u_\theta\bar{e}_\theta)$ is $r u_\theta$ (from $\bar{e}_r\wedge\bar{e}_\theta = \bar{e}_z$). The antiderivative of r^2 is $r^3/3$; Barrow's rule between R_i and R_o gives $(R_o^3 - R_i^3)/3$.

$$\boxed{M = \frac{2\pi}{3}\,\rho_1 u_1 u_\theta\left(R_o^3 - R_i^3\right)} \tag{3}$$

Total energy (Notes Eq. 3.46)

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho\left(e + \tfrac{v^2}{2}\right)dV}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\left(e + \frac{v^2}{2}\right)\left[(\bar{v} - \underbrace{\bar{v}_c}_{0})\cdot\bar{n}\right]d\sigma = -\int_{\Sigma_c} p\,\bar{v}\cdot\bar{n}\,d\sigma + \int_{\Sigma_c} \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma + \underbrace{\int_{V_c}\rho\bar{g}\cdot\bar{v}\,dV}_{\text{neglect }\bar{g}} - \int_{\Sigma_c}\bar{q}\cdot\bar{n}\,d\sigma + \underbrace{\int_{V_c} Q_R\,dV}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho \left(e + \frac{v^2}{2}\right) (\bar{v} \cdot \bar{n})\, d\sigma$	$-\int p\, \bar{v} \cdot \bar{n}\, d\sigma$	$\int \bar{v} \cdot (\bar{\bar{\tau}}' \cdot \bar{n})\, d\sigma$	$-\int \bar{q} \cdot \bar{n}\, d\sigma$
Σ_{in}	$-G\left(e_1 + \frac{u_1^2+u_\theta^2}{2}\right)$	$+p_1 u_1 \pi (R_o^2 - R_i^2)$	0	0
Σ_{out}	$+G\left(e_2 + \frac{u_2^2}{2}\right)$	$-p_2 u_2 \pi R_o^2$	0	0
$\Sigma_w,\ \Sigma_a$	0	0 ($\bar{v} = 0$)	0 ($\bar{v} = 0$)	0 (adiabatic)

$$\boxed{\frac{\gamma}{\gamma - 1}\frac{p_1}{\rho_1} + \frac{u_1^2 + u_\theta^2}{2} = \frac{\gamma}{\gamma - 1}\frac{p_2}{\rho_2} + \frac{u_2^2}{2}} \qquad (4)$$

(1) and (4) give ρ_2 and u_2; then (2) gives F and (3) gives M directly; $T_2 = p_2/(\rho_2 R_g)$ from the equation of state.

Discrepancy with the official solution (sign of the torque)

CL14.pdf uses $\bar{x} \wedge \bar{e}_\theta = -r\bar{e}_z$ and obtains $M = -\frac{2\pi}{3}\rho_1 u_1 u_\theta (R_e^3 - R_i^3)$ (it calls the outer radius R_e). With the right-handed basis $(\bar{e}_r, \bar{e}_\theta, \bar{e}_z)$ one has $\bar{e}_r \wedge \bar{e}_\theta = +\bar{e}_z$, so $\bar{x} \wedge \bar{e}_\theta = +r\bar{e}_z$ and the sign is the one in (3). The magnitude is the same; physically, the torque of the gas on the stator has the sense of the incoming swirl, since the stator removes it. The computed sign is kept.

PHASE 4

Interpretation, Limits and Dimensional Check

- (4): the stagnation enthalpy is conserved; the kinetic energy of the swirl, $u_\theta^2/2$, is recovered as enthalpy (pressure and temperature) downstream.
- For $u_\theta = 0$ there is nothing to remove and $M = 0$. M is the torque that the mounting of the stator must withstand.
- Dimensions (SI): $[(p + \rho u^2)R^2] = \mathrm{N}$; $[\rho u_1 u_\theta R^3] = \mathrm{N \cdot m}$.
- Numerical illustration (hypothetical data, extension): $p_1 = 1.2 \times 10^5$ Pa, $\rho_1 = 1.1$ kg/m^3, $u_1 = 150$ m/s, $u_\theta = 80$ m/s, $p_2 = 1.3 \times 10^5$ Pa, $R_i = 0.2$ m, $R_o = 0.5$ m, $\gamma = 1.4$: eliminating $\rho_2 = G/(\pi R_o^2 u_2)$ in (4) gives a quadratic in u_2 with positive root $u_2 = 118.57$ m/s, $\rho_2 = 1.1689$ kg/m^3; then $F = -19512$ N (the module is pushed upstream by the pressure rise) and $M = 3234.6$ N·m.

PROBLEM CL15

Mixing Elbow: Exit Velocity and Temperature, Force and Torque

Original statement · CL15.pdf

Two inlets | Mixing | Energy | Angular momentum

OFFICIAL PROBLEM STATEMENT

Consider the mixing elbow shown schematically in the figure of CL15.pdf. Cold water flows through the main pipe and mixes with warmer water that enters through a smaller inlet located at the elbow. The liquid properties can be considered independent of the temperature and gravity effects can be neglected. All pipe walls are adiabatic. The known pipe dimensions, fluid properties and boundary conditions are given in the figure (density ρ, viscosity μ, conductivity k, specific heat c; main inlet v_1, p_1, T_1, diameter D_1; secondary inlet v_2, p_2, T_2, diameter D_2; exit at p_a). Assume that the two streams mix completely and that the velocity and temperature are uniform at the exit section.

- Find the velocity v_s and temperature T_s of the mixed stream emerging from the pipe at ambient pressure.
- Obtain the net force $\bar{F}$ exerted on the mixing elbow by the water flowing inside and the ambient air.
- Obtain the torque $\bar{M}_1$ about the point $\bar{x}_1$ located at the center of the larger inlet.
- Compute the above results numerically, given the following data: $v_1 = 0.4$ m/s; $v_2 = 1.2$ m/s; $\rho = 1000$ kg/m^3; $T_1 = 20^\circ$C; $T_2 = 40^\circ$C; $\mu = 8 \times 10^{-4}$ Pa s; $p_1 - p_a = 50$ Pa; $p_2 - p_a = 200$ Pa; $k = 0.677$ W/(m s); $D_1 = 10$ cm; $D_2 = 2.5$ cm; $c = 4216$ J/(kg K); $L_1 = 20$ cm; $L_2 = 7$ cm.

Source: CL15.pdf (pages 1-2, exam of Dec 18, 2012, with solution). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral angular momentum balance Eq. 3.38, integral total energy balance Eq. 3.46.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, incompressible liquid with constant properties, $e = e_0 + cT$; neglect $\bar{g}$; adiabatic walls; uniform properties at the three sections; complete mixing.
- Geometry (figure): main inlet along $+\bar{e}_x$ (section $A_1 = \pi D_1^2/4$), secondary inlet along $+\bar{e}_y$ ($A_2 = \pi D_2^2/4$), exit along $+\bar{e}_y$ (section A_1) at p_a. The secondary inlet and the exit share the same vertical axis, at a horizontal distance $L_1 + D_1 + D_1/2 = L_1 + \frac{3}{2}D_1$ from $\bar{x}_1$ (bend of inner radius D_1; reading of the figure, as in the official solution).
- Data: as listed. Unknowns: $[v_s,\ T_s,\ \bar{F},\ \bar{M}_1]$. μ, k and L_2 are not needed.

Symbol	Meaning	SI unit
v_1, v_2, v_s	inlet and exit velocities	m/s
T_1, T_2, T_s	temperatures	°C
$p_1 - p_a, p_2 - p_a$	inlet gauge pressures	Pa
D_1, D_2, L_1	diameters and length	m
$\bar{F}, \bar{M}_1$	force and torque on the elbow	N, N·m

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the elbow: $\bar{e}_x$ horizontal (main inlet flow), $\bar{e}_y$ vertical upwards, $\bar{e}_z$ towards the reader; moments about $\bar{x}_1$.
- Surfaces (fixed CV, as in the official solution): Σ_1 (main inlet, $\bar{n} = -\bar{e}_x$), Σ_2 (secondary inlet, $\bar{n} = -\bar{e}_y$), Σ_{out} (exit, $\bar{n} = +\bar{e}_y$), Σ_{wall}.

$$\boxed{\bar{F} = -\int_{\Sigma_{wall}} \left[-(p - p_a)\,\bar{n} + \bar{\bar{\tau}}' \cdot \bar{n}\right] d\sigma}$$

$$\boxed{\bar{M}_1 = -\int_{\Sigma_{wall}} (\bar{x} - \bar{x}_1) \wedge \left[-(p - p_a)\,\bar{n} + \bar{\bar{\tau}}' \cdot \bar{n}\right] d\sigma}$$

PHASE 3

Balances, Surface by Surface

Mass (Notes Eq. 3.11)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c} \rho\, dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\,(\bar{v} - \underbrace{\cancel{\bar{v}_c}}_{0}) \cdot \bar{n}\, d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\, \bar{v} \cdot \bar{n}\, d\sigma$
Σ_1	$-\rho v_1 A_1$
Σ_2	$-\rho v_2 A_2$
Σ_{out}	$+\rho v_s A_1$
Σ_{wall}	0

$$v_1 A_1 + v_2 A_2 = v_s A_1 \implies \boxed{v_s = v_1 + v_2 \left(\frac{D_2}{D_1}\right)^2} \qquad (1)$$

Total energy (Notes Eq. 3.46)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c} \rho\left(e + \frac{v^2}{2}\right) dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\left(e + \frac{v^2}{2}\right)\left[(\bar{v} - \underbrace{\cancel{\bar{v}_c}}_{0}) \cdot \bar{n}\right] d\sigma = -\int_{\Sigma_c} p\,\bar{v} \cdot \bar{n}\, d\sigma + \int_{\Sigma_c} \bar{v} \cdot (\bar{\bar{\tau}}' \cdot \bar{n})\, d\sigma + \underbrace{\int_{V_c} \rho \bar{g} \cdot \cancel{\bar{v}\, dV}}_{\text{neglect } \bar{g}} - \int_{\Sigma_c} \bar{q} \cdot \bar{n}\, d\sigma + \underbrace{\cancel{\int_{V_c} Q_H\, dV}}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho \left(e + \frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	$-\int \bar{q}\cdot\bar{n}\,d\sigma$
Σ_1	$-\rho\left(e_1+\frac{v_1^2}{2}\right)v_1A_1$	$+p_1v_1A_1$	0	0
Σ_2	$-\rho\left(e_2+\frac{v_2^2}{2}\right)v_2A_2$	$+p_2v_2A_2$	0	0
Σ_{out}	$+\rho\left(e_s+\frac{v_s^2}{2}\right)v_sA_1$	$-p_av_sA_1$	0	0
Σ_{wall}	0	0	0	0 (adiabatic)

With $e = e_0 + cT$, dividing by $\rho c\, v_s A_1$ (e_0 cancels by (1)) and writing p_a as $p_a(v_1A_1 + v_2A_2)/(v_sA_1)$:

$$\boxed{T_s = \left(T_1 + \frac{p_1 - p_a}{\rho c} + \frac{v_1^2}{2c}\right)\frac{v_1}{v_s} + \left(T_2 + \frac{p_2 - p_a}{\rho c} + \frac{v_2^2}{2c}\right)\frac{v_2A_2}{v_sA_1} - \frac{v_s^2}{2c}} \quad (2)$$

`Linear momentum (Notes Eq. 3.36)`

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\bar{v}\,[(\bar{v} - \underbrace{\bar{v}_c}_{0})\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c}(p - p_a)\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int (p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_1	$-\rho v_1^2 A_1\,\bar{e}_x$	$(p_1 - p_a)A_1\,\bar{e}_x$	0
Σ_2	$-\rho v_2^2 A_2\,\bar{e}_y$	$(p_2 - p_a)A_2\,\bar{e}_y$	0
Σ_{out}	$+\rho v_s^2 A_1\,\bar{e}_y$	$0\ (p = p_a)$	0
Σ_{wall}	0	$-\int (p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed terms: $-\bar{F}$.

$$\boxed{\bar{F} = \left(\rho v_1^2 + p_1 - p_a\right)A_1\,\bar{e}_x + \left[\left(\rho v_2^2 + p_2 - p_a\right)A_2 - \rho v_s^2 A_1\right]\bar{e}_y} \quad (3)$$

Angular momentum about $\bar{x}_1$ (Notes Eq. 3.38)

$$\underbrace{\frac{d}{dt}\int_{V_c}\rho\,(\bar{x}-\bar{x}_1)\wedge\bar{v}\,dV}_{0}+\int_{\Sigma_c}\rho\,[(\bar{x}-\bar{x}_1)\wedge\bar{v}](\bar{v}\cdot\bar{n})\,d\sigma=-\int_{\Sigma_c}(\bar{x}-\bar{x}_1)\wedge(p-p_a)\bar{n}\,d\sigma+\int_{\Sigma_c}(\bar{x}-\bar{x}_1)\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int\rho\,[(\bar{x}-\bar{x}_1)\wedge\bar{v}](\bar{v}\cdot\bar{n})\,d\sigma$	$-\int(\bar{x}-\bar{x}_1)\wedge(p-p_a)\bar{n}\,d\sigma$	$\int(\bar{x}-\bar{x}_1)\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$
Σ_1	0 (centred at $\bar{x}_1$)	0	0
Σ_2	$-\rho v_2^2 A_2\,d\,\bar{e}_z$	$(p_2-p_a)A_2\,d\,\bar{e}_z$	0
Σ_{out}	$+\rho v_s^2 A_1\,d\,\bar{e}_z$	0	0
Σ_{wall}	0	$-\int(\bar{x}-\bar{x}_1)\wedge(p-p_a)\bar{n}\,d\sigma$	$\int(\bar{x}-\bar{x}_1)\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$

Boxed terms: $-\bar{M}_1$. With $d=L_1+\frac{3}{2}D_1$ the horizontal distance to the common vertical axis, $(\bar{x}-\bar{x}_1)\wedge\bar{e}_y=d\,\bar{e}_x\wedge\bar{e}_y=d\,\bar{e}_z$ for the centres of Σ_2 and Σ_{out} (the vertical offsets are parallel to $\bar{e}_y$ and drop out).

$$\boxed{\bar{M}_1=\left[\left(\rho v_2^2+p_2-p_a\right)A_2-\rho v_s^2 A_1\right]\left(L_1+\frac{3}{2}D_1\right)\bar{e}_z}\qquad(4)$$

Numerical values (part 4)

Quantity	**Value**
v_s	0.475 m/s
T_s	23.16 °C
$\bar{F}$	$1.65\,\bar{e}_x-0.967\,\bar{e}_y$ N
$\bar{M}_1$	$-0.338\,\bar{e}_z$ N·m

PHASE 4

Interpretation, Limits and Dimensional Check

- (2) is a mass-weighted average of the inlet temperatures, corrected by the tiny flow-work and kinetic-energy terms ($\Delta p/(\rho c)\sim 10^{-5}$ K here): $T_s\approx(v_1A_1T_1+v_2A_2T_2)/(v_sA_1)$.

- $F_y < 0$: the exit momentum flux exceeds the momentum and gauge pressure brought in by the small inlet, so the elbow is pulled downwards; $\bar{M}_1$ follows from the same vertical imbalance acting at the arm d.
- Dimensions (SI): $[(p - p_a)/(\rho c)] = \mathrm{K}$, $[v^2/c] = \mathrm{K}$; $[\rho v^2 A] = \mathrm{N}$; $[\rho v^2 A d] = \mathrm{N \cdot m}$.
- Agreement with the official solution: CL15.pdf obtains (1) to (4) and $v_s = 0.475$ m/s, $T_s = 23^\circ$C, $\bar{F} = 1.65\,\bar{e}_x - 0.967\,\bar{e}_y$ N, $\bar{M}_1 = -0.338\,\bar{e}_z$ N·m, the same values.

PROBLEM CL16

Plane Jet Split by a Non-Symmetric Wedge: Force, Moment and Heat

Original statement · CL16.pdf

Free jets | Bernoulli | Moment of a jet | Heat transfer

OFFICIAL PROBLEM STATEMENT

A plane liquid jet of constant density ρ, thickness h and uniform velocity v, impinges on a non-symmetric wedge of geometry and dimensions shown in the figure of CL16.pdf. The wedge splits the jet into two layers of unequal flux, such that a portion α of the volume flux is deflected an angle θ, and the rest of the fluid is deflected an angle 2θ. The flow is steady; viscous and gravitational forces are negligible; the known data are $(\rho, c, v, h, \alpha, \theta, L, p_a, T_0, T_1, T_2)$.

- Noting that the conditions for the use of Bernoulli's theorem are met, determine the (uniform) velocity of the fluid on each deflected layer.
- Determine the thickness h_1 and h_2 of each layer.
- Determine the force $\bar{F}$ (per unit length in the z-direction) that the liquid jet and the ambient air exert on the wedge.
- Determine the moment $\bar{M}_0$ due to the force $\bar{F}$ about the corner $\bar{x}_0$ of the wedge. Find the value of the fraction α that would produce a null moment, $\bar{M}_0 = 0$.
- Knowing the uniform temperature T_0 of the incoming jet, the uniform temperatures T_1 and T_2 of the two liquid layers when they lose contact with the wedge, and the constant specific heat c of the liquid, determine the heat q transferred per unit surface from the liquid to the wedge. Assume that the heat flux is uniform on the wedge surface and that there is no heat transfer by conduction between the liquid and the ambient air.

Source: CL16.pdf (pages 1-2, exam of Jan 14, 2013, with handwritten results). Notes.pdf: Bernoulli's equation Eq. 4.17, integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral angular momentum balance Eq. 3.38, integral total energy balance Eq. 3.46.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, planar (per unit length in z), incompressible, frictionless; neglect $\bar{g}$; free jets at p_a (straight parallel streamlines in the layers away from the corner, so $p = p_a$ inside them).

- Geometry (figure): the incoming jet is centred on the x axis, which passes through the corner $\bar{x}_0$; layer 1 leaves along $\bar{t}_1 = \cos\theta\,\bar{e}_x + \sin\theta\,\bar{e}_y$ on the upper face, layer 2 along $\bar{t}_2 = \cos 2\theta\,\bar{e}_x - \sin 2\theta\,\bar{e}_y$ on the lower face. The faces have horizontal length L, i.e. lengths $L/\cos\theta$ and $L/\cos 2\theta$.
- Heat: uniform flux q from the liquid to the wedge on both faces; no conduction to the air; $e = cT$.
- Data: $[\rho,\ c,\ v,\ h,\ \alpha,\ \theta,\ L,\ p_a,\ T_0,\ T_1,\ T_2]$. Unknowns: $[v_1,\ v_2,\ h_1,\ h_2,\ \bar{F},\ \bar{M}_0,\ q]$.

Symbol	Meaning	SI unit
v, h	jet velocity and thickness	m/s, m
α, θ	flux fraction and deflection angle	1, rad
h_1, h_2	layer thicknesses	m
$\bar{F}$, $\bar{M}_0$	force and moment per unit length	N/m, N·m/m
q	heat flux to the wedge	W/m^2

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the wedge; origin at the corner $\bar{x}_0$; $\bar{e}_z$ towards the reader.
- Surfaces: Σ_0 (incoming jet section, $\bar{n} = -\bar{e}_x$), Σ_1, Σ_2 (layer sections normal to $\bar{t}_1, \bar{t}_2$, $\bar{n} = \bar{t}_1, \bar{t}_2$), Σ_a (free surfaces, $p = p_a$) and Σ_w (wetted faces). On layer k, $\bar{x} - \bar{x}_0 = s\,\bar{t}_k + \eta\,\bar{m}_k$, with $\bar{m}_1 = -\sin\theta\,\bar{e}_x + \cos\theta\,\bar{e}_y$, $\bar{m}_2 = -\sin 2\theta\,\bar{e}_x - \cos 2\theta\,\bar{e}_y$ the unit normals pointing away from the wedge and $0 \le \eta \le h_k$.

$$\boxed{\bar{F} = \int_{\Sigma_w} \left[(p - p_a)\,\bar{n} - \bar{\bar{\tau}}' \cdot \bar{n}\right] d\sigma}$$

$$\boxed{\bar{M}_0 = \int_{\Sigma_w} (\bar{x} - \bar{x}_0) \wedge \left[(p - p_a)\,\bar{n} - \bar{\bar{\tau}}' \cdot \bar{n}\right] d\sigma}$$

$$\boxed{q = -\frac{1}{L/\cos\theta + L/\cos 2\theta} \int_{\Sigma_w} \bar{q} \cdot \bar{n}_{\text{in}}\, d\sigma}$$

PHASE 3

Bernoulli and Balances, Surface by Surface

1. Velocities

Bernoulli (Notes Eq. 4.17, $U = 0$) along a free streamline from the incoming jet to each layer, with $p = p_a$ at both ends: $p_a + \frac{1}{2}\rho v^2 = p_a + \frac{1}{2}\rho v_k^2$, so

$$\boxed{v_1 = v_2 = v} \tag{1}$$

2. Mass (Notes Eq. 3.11): thicknesses

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho\, dV}}_{0\ (\text{steady})} + \int_{\Sigma_c} \rho\,(\bar{v} - \underbrace{\cancel{\bar{v}_c}}_{0}) \cdot \bar{n}\, d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\, \bar{v} \cdot \bar{n}\, d\sigma$
Σ_0	$-\rho v h = -\rho Q$
Σ_1	$+\rho v h_1 = \rho \alpha Q$
Σ_2	$+\rho v h_2 = \rho(1-\alpha) Q$
$\Sigma_a,\ \Sigma_w$	0

$$\boxed{h_1 = \alpha h, \qquad h_2 = (1-\alpha) h} \tag{2}$$

3. Linear momentum (Notes Eq. 3.36)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho\bar{v}\,dV}}_{0\ (\text{steady})}+\int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}]\,d\sigma=-\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma+\int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma+\underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int(p-p_a)\,\bar{n}\,d\sigma$	$\int\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_0	$-\rho v^2 h\,\bar{e}_x$	0	0
Σ_1	$\rho v^2\alpha h\,(\cos\theta\,\bar{e}_x+\sin\theta\,\bar{e}_y)$	0	0
Σ_2	$\rho v^2(1-\alpha)h\,(\cos 2\theta\,\bar{e}_x-\sin 2\theta\,\bar{e}_y)$	0	0
Σ_a	0	$0\ (p=p_a)$	0
Σ_w	0	$-\int_{\Sigma_w}(p-p_a)\,\bar{n}\,d\sigma$	$\int_{\Sigma_w}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed Σ_w terms: $-\bar{F}$.

$$\boxed{\bar{F}=\rho v^2 h\,\{[1-\alpha\cos\theta-(1-\alpha)\cos 2\theta]\,\bar{e}_x-[\alpha\sin\theta-(1-\alpha)\sin 2\theta]\,\bar{e}_y\}}\quad(3)$$

4. Angular momentum about $\bar{x_0}$ (Notes Eq. 3.38)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho\,(\bar{x}-\bar{x}_0)\wedge\bar{v}\,dV}}_{0}+\int_{\Sigma_c}\rho\,[(\bar{x}-\bar{x}_0)\wedge\bar{v}](\bar{v}\cdot\bar{n})\,d\sigma=-\int_{\Sigma_c}(\bar{x}-\bar{x}_0)\wedge(p-p_a)\bar{n}\,d\sigma+\int_{\Sigma_c}(\bar{x}-\bar{x}_0)\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int\rho\,[(\bar{x}-\bar{x}_0)\wedge\bar{v}](\bar{v}\cdot\bar{n})\,d\sigma$	pressure and viscous moments
Σ_0	$\int_{-h/2}^{h/2}\rho v\,(-yv)(-1)\,dy\,\bar{e}_z=0$ (odd in y)	0
Σ_1	$\int_0^{h_1}\rho v^2\,\eta\,(\bar{m}_1\wedge\bar{t}_1)\,d\eta=-\rho v^2\frac{h_1^2}{2}\,\bar{e}_z$	0
Σ_2	$\int_0^{h_2}\rho v^2\,\eta\,(\bar{m}_2\wedge\bar{t}_2)\,d\eta=+\rho v^2\frac{h_2^2}{2}\,\bar{e}_z$	0
Σ_w	0	$-\bar{M}_0$

Here $\bar{m}_1 \wedge \bar{t}_1 = (-\sin^2\theta - \cos^2\theta)\,\bar{e}_z = -\bar{e}_z$ and $\bar{m}_2 \wedge \bar{t}_2 = (\sin^2 2\theta + \cos^2 2\theta)\,\bar{e}_z = +\bar{e}_z$; the part $s\,\bar{t}_k \wedge \bar{t}_k$ vanishes. The antiderivative of η is $\eta^2/2$; Barrow's rule between 0 and h_k gives $h_k^2/2$. With (2): $-\frac{\rho v^2 h^2}{2}\left[\alpha^2 - (1-\alpha)^2\right] = -\frac{\rho v^2 h^2}{2}(2\alpha - 1)$.

$$-\frac{\rho v^2 h^2}{2}(2\alpha-1)\,\bar{e}_z = -\bar{M}_0 \implies \boxed{\bar{M}_0 = \rho v^2\,\frac{h^2}{2}\,(2\alpha-1)\,\bar{e}_z, \qquad \bar{M}_0 = 0 \iff \alpha = \frac{1}{2}} \quad (4)$$

5. Total energy (Notes Eq. 3.46)

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho\left(e + \tfrac{v^2}{2}\right) dV}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\left(e + \frac{v^2}{2}\right)\left[(\bar{v} - \underbrace{\bar{v}_c}_{0})\cdot\bar{n}\right] d\sigma = -\int_{\Sigma_c} p\,\bar{v}\cdot\bar{n}\,d\sigma + \int_{\Sigma_c} \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma + \underbrace{\int_{V_c} \rho\bar{g}\cdot\bar{v}\,dV}_{\text{neglect } \bar{g}} - \int_{\Sigma_c} \bar{q}\cdot\bar{n}\,d\sigma + \underbrace{\int_{V_c} Q_H\,dV}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e + \frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	$-\int \bar{q}\cdot\bar{n}\,d\sigma$
Σ_0	$-\rho v h\left(cT_0 + \frac{v^2}{2}\right)$	$+p_a v h$	0	0
Σ_1	$+\rho v\alpha h\left(cT_1 + \frac{v^2}{2}\right)$	$-p_a v\alpha h$	0	0
Σ_2	$+\rho v(1-\alpha) h\left(cT_2 + \frac{v^2}{2}\right)$	$-p_a v(1-\alpha)h$	0	0
Σ_a	0	0	0	0 (no conduction to the air)
Σ_w	0	0 ($\bar{v}=0$)	0	$-q\left(\frac{L}{\cos\theta} + \frac{L}{\cos 2\theta}\right)$

Boxed Σ_w term: the heat leaves the liquid towards the wedge. The kinetic-energy and pressure-work terms cancel because $\alpha + (1-\alpha) = 1$.

$$\rho v h c\,[\alpha T_1 + (1-\alpha)T_2 - T_0] = -q\left(\frac{L}{\cos\theta} + \frac{L}{\cos 2\theta}\right) \implies \boxed{q = \frac{\rho c v h}{\frac{L}{\cos\theta} + \frac{L}{\cos 2\theta}}\,[T_0 - \alpha T_1 - (1-\alpha)T_2]} \quad (5)$$

PHASE 4

Interpretation, Limits and Dimensional Check

- Symmetric-split check: with $\alpha = 1/2$ the two layers carry equal flux and their moments about the corner cancel, (4). The force (3) is still not symmetric because the deflections θ and 2θ differ.
- Limit $\theta \to 0$: no deflection, $\bar{F} \to 0$ in (3) (both brackets vanish).
- (5): $q > 0$ (liquid heats the wedge) when the liquid leaves colder than it arrived on average.
- Dimensions (SI, per unit length): $[\rho v^2 h] = \mathrm{N/m}$; $[\rho v^2 h^2] = \mathrm{N \cdot m/m}$; $[\rho c v h\, \Delta T / L] = \mathrm{W/m^2}$.
- Numerical cross-check of (4) (hypothetical data): $\rho = 1000$ kg/m^3, $v = 10$ m/s, $h = 0.02$ m, $\alpha = 0.3$, $\theta = 25^\circ$: midpoint quadrature of the two layer integrals gives an angular-momentum flux of 8.000 N·m/m, equal to $-\rho v^2 h^2 (2\alpha - 1)/2 = 8$, i.e. $M_{0,z} = -8$ N·m/m.
- Agreement with the official solution: CL16.pdf obtains (1) to (5).